

A Cheatsheet for Lie Algebra

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These notes follow Humphreys, *Introduction to Lie Algebras and Representation Theory* [1]. Throughout, F denotes an algebraically closed field.

1 Basic Concepts

1.1 Preliminaries

Definition 1.1. A vector space L over a field F with an operation $L \times L \rightarrow L$
 $(x, y) \mapsto [xy]$
called the bracket or commutator of x, y is called a Lie algebra over F if the following axioms are satisfied $\left\{ \begin{array}{l} \text{the bracket operator is bilinear} \\ \forall x \in L, [xx] = 0 \text{ where the last} \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = 0 \end{array} \right.$
condition is called the Jacobi identity.

$$\begin{aligned}
& [x+y, x+y] = [x, x] + [x, y] + [y, x] + [y, y] \quad \text{by the linearity} \\
\rightarrow & \quad \quad \quad = [x, y] + [y, x] \quad [x, x] = 0, \forall x \in L \Leftrightarrow [x, y] = -[y, x] \\
& \quad \quad \quad = 0 \quad [x, x] = 0, \forall x \in L
\end{aligned}$$

and if $\text{char } F \neq 2$, the converse also holds.

Definition 1.2. Two Lie algebras L, L' over F are isomorphic if there exists a vector space isomorphism $\phi: L \rightarrow L'$ satisfying $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$.

Definition 1.3. A subalgebra of a Lie algebra L is a subspace K of L s.t. $[xy] \in K$, $\forall x, y \in K$.

$\rightarrow K$ is a Lie algebra with the inherited operations.

e.g. **The General Linear Algebra** $\mathfrak{gl}(V)$

If V is a finite dimensional vector space over F , the set of linear transformations $V \rightarrow V$ is denoted by $\text{End}V$, which is of dimension n^2 where $\dim V = n$ and is a ring relative to the usual product operation. Define a new operation $[x, y] = xy - yx$, called the bracket of x, y . With this operation, $\text{End}V$ becomes a Lie algebra over F .

$\rightarrow \begin{cases} \forall x \in L, [x, x] = xx - xx = 0 \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = x(yz - zy) + y(zx - xz) + z(xy - yx) = 0 \end{cases}$
This Lie algebra is called the general linear algebra $\mathfrak{gl}(V)$. It is identified with the set of all $n \times n$ matrices over F , denoted $\mathfrak{gl}(n, F)$. Then $[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{li}e_{kj}$ where e_{ij} is a standard basis for matrices.

Subalgebras of the General Linear Algebra

- **the Special Linear Algebra** is the set of endomorphisms of V having trace zero and is denoted $\mathfrak{sl}(V) = \mathfrak{sl}(l+1, F)$ where $\dim V = l+1$.

Throughout, $T(\cdot)$ denotes the trace. Since $\begin{cases} T(xy) = T(yx) \\ T(x+y) = T(x) + T(y) \end{cases}$, it is a subalgebra of $\mathfrak{gl}(V)$.

Take all e_{ij} , $i \neq j$ along with $h_i = e_{ii} - e_{i+1, i+1}$. Then this is the basis for $\mathfrak{sl}(V)$ and therefore it has dimension $l + (l+1)^2 - (l+1)$.

- **the symplectic algebra** is the set of endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$, denoted $\mathfrak{sp}(V) = \mathfrak{sp}(2l, F)$, where $\dim V = 2l$ and f has a skew-symmetric form $s = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Then for $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$ where $m, n, p, q \in \mathfrak{gl}(l, F)$, $sx = -x's$ given it being symplectic. In other words, $\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} =$

$$-\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \Leftrightarrow \begin{cases} n' = n \\ p' = p \\ m' = -q \end{cases} . \text{ Then the basis consists of}$$

$$\begin{cases} m' = -q & \begin{cases} e_{ii} - e_{l+i, l+i}, & 1 \leq i \leq l \\ e_{ij} - e_{l+j, l+i} & l + l^2 - l \end{cases} \\ n = n' & \begin{cases} e_{i, l+i} \\ e_{i, l+j} + e_{j, l+i} & l + \frac{1}{2}(l(l-1)) \end{cases} \end{cases} .$$
 Since n and p share the basis, the total dimension becomes $2l^2 + l$.

- the Orthogonal Algebra** is the set of endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$, denoted $\mathfrak{o}(V) = \mathfrak{o}(2l, F)$ where $\dim V = 2l$ and f has a form $s = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$. Then for $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$ where $m, n, p, q \in \mathfrak{gl}(l, F)$, $sx = -x's$

given it being orthogonal. In other words, $\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} = -\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$.

Then the basis consists of

$$\begin{cases} m' = -q & \text{similar with the symplectic case} & l^2 \\ n = -n' & \begin{cases} \text{diagonal} = 0 \\ e_{i, l+j} - e_{j, l+i} & \frac{1}{2}(l(l-1)) \end{cases} \end{cases} .$$

Since n and p share the basis, the total dimension becomes $2l^2 - l$.

- $$\begin{cases} \text{the set of upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i > j & \mathfrak{t}(n, F) \\ \text{the set of strictly upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i \geq j & \mathfrak{n}(n, F) \\ \text{the set of all diagonal matrices} & \mathfrak{d}(n, F) \end{cases} \rightarrow$$

$\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$

pf. The basis for $\mathfrak{t}(n, F)$ consists of all e_{ij} s.t. $i \leq j$ which has dimension $\frac{1}{2}n(n+1)$. The basis for $\mathfrak{d}(n, F)$ consists of all e_{ii} which has dimension n , while the basis for $\mathfrak{n}(n, F)$ consists of all e_{ij} s.t. $i < j$ which has dimension $\frac{1}{2}n(n-1)$. Then the sum of bases for $\mathfrak{n}(n, F)$, $\mathfrak{d}(n, F)$ gives the basis for $\mathfrak{t}(n, F)$ and the stmt follows. Moreover the sum is direct: $\mathfrak{n}(n, F)$ is nonabelian while $\mathfrak{d}(n, F)$ is abelian, and their bases are disjoint (e_{ij} with $i < j$ versus e_{ii}), so $\mathfrak{n}(n, F) \cap \mathfrak{d}(n, F) = 0$ and $\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$.

Definition 1.4. For any field F , an F -algebra is a vector space $\mathcal{U}$ over F endowed with a bilinear operation $\mathcal{U} \times \mathcal{U} \rightarrow \mathcal{U}$, usually denoted by juxtaposition. By a derivation of $\mathcal{U}$, we mean a linear map $\delta : \mathcal{U} \rightarrow \mathcal{U}$ satisfying the product rule $\delta(ab) = a\delta(b) + \delta(a)b$.

Since a Lie algebra L is an F -algebra in the above sense, $\text{Der}L$ is defined. If $x \in L$, $y \mapsto [xy]$ is an endomorphism of L , denoted adx . Then $\text{adx} \in \text{Der}L$ by the Jacobi identity. Derivations of this form are called **inner**, all others **outer**.

Definition 1.5. The map $\begin{matrix} L & \rightarrow & \text{Der}L \\ x & \mapsto & \text{adx} \end{matrix}$ is called the adjoint representation of L .

Definition 1.6. A Lie algebra L over F is called abelian if $\forall x, y \in L, [xy] = 0$.

Definition 1.7. A subspace I of a Lie algebra L is called an ideal of L if $x \in L, y \in I$ imply $[xy] \in I$.

$\rightarrow [yx] = -[xy] \in I$ by skew-symmetry.

e.g. $\begin{cases} Z(L) = \{z \in L | [xz] = 0, \forall x \in L\}, \text{ the centre of } L \\ [LL] \text{ the derived algebra of } L \end{cases}$

Then L is abelian iff $[LL] = 0$. At the other extreme, when $L = \mathfrak{sl}(n, F)$, $[LL] = L$.

If I, J are two ideals of a Lie algebra L , then $\begin{cases} I + J = \{x + y | x \in I, y \in J\} \\ [IJ] = \{\sum [x_i y_i] | x_i \in I, y_i \in J\} \end{cases}$ are ideals.

Definition 1.8. A Lie algebra L is called simple if it has no ideals except itself and 0, and $[LL] \neq 0$.

e.g. $L = \mathfrak{sl}(2, F)$, $\text{char } F \neq 2$. Then it has a standard basis $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then the multiplication table is determined by the equations $\begin{cases} [xy] = h \\ [hx] = 2x \\ [hy] = -2y \end{cases}$. If $I \neq 0$ is an ideal of L , let $ax + by + ch$ be an arbitrary

nonzero element of I . Then applying $\begin{cases} \text{adx twice, we get } -2bx \in I \\ \text{ady twice, we get } -2ay \in I \end{cases}$. If a or b is nonzero, that means either x or y is in I , and therefore $L = I$ by the above multiplication table. If $a = b = 0$, $ch \in I$ and this also implies $L = I$ by the above multiplication table. This shows L is simple.

If L is not simple, it is possible to factor out a nonzero proper ideal I and obtain a **quotient algebra** L/I : as vector space L/I is just the quotient space, while its Lie multiplication is defined by $[x + I, y + I] = [xy] + I$.

Definition 1.9. For any Lie algebras L, L' over F , a linear transformation $\phi : L \rightarrow L'$ is called a homomorphism if $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$. In addition, if $\ker \phi = 0$, ϕ is called a monomorphism, or if $\text{Im } \phi = L'$, ϕ is called an epimorphism. It is an isomorphism if it is both mono- and epi-.

$\ker \phi$ is an ideal of L .

pf. $\forall x \in \ker \phi, \forall y \in L, \phi([xy]) = [\phi(x)\phi(y)] = 0$ so $[xy] \in \ker \phi$ and therefore the stmt holds.

$\text{Im } \phi$ is a subalgebra of L' .

Proposition 1.1. 1. If $\phi : L \rightarrow L'$ is a homomorphism of Lie algebras, then $L/\ker \phi \cong \text{Im} \phi$. If I is any ideal of L included in $\ker \phi$, there exists a unique

homomorphism $\psi : L/I \rightarrow L'$ making the following diagram commute:

$$\begin{array}{ccc} L & \xrightarrow{\phi} & L' \\ & \searrow \pi & \uparrow \psi \\ & & L/I \end{array}$$

2. If I, J are ideals of L s.t. $I \subset J$, then J/I is an ideal of L/I and $(L/I)/(J/I)$ is naturally isomorphic to L/J .

3. If I, J are ideals of L , there is a natural isomorphism between $(I + J)/J$ and $I/(I \cap J)$.

Definition 1.10. A representation of a Lie algebra L is a homomorphism $\phi : L \rightarrow \mathfrak{gl}(V)$.

For L finite-dimensional, we have an adjoint representation $\text{ad} : L \rightarrow \mathfrak{gl}(L)$. Then its kernel consists of all $x \in L$ for which $\text{adx} = 0$, i.e. $[xy] = 0$. So $\ker \text{ad} = Z(L)$. Then, if L is simple, $Z(L) = 0$ so that $\text{ad}L : L \rightarrow \mathfrak{gl}(L)$ is a monomorphism. This means that any simple Lie algebra is isomorphic to a linear Lie algebra.

Definition 1.11. An automorphism of L is an isomorphism of L onto itself. The group of automorphisms of L is denoted $\text{Aut}L$.

If $g \in \text{GL}(V)$ is any invertible endomorphism of V and if moreover $gLg^{-1} = L$, then $x \mapsto gxg^{-1}$ is an automorphism of L .

Assume $\text{char} F = 0$. Suppose $x \in L$ is an element for which adx is nilpotent, i.e. $(\text{adx})^k = 0$, $\exists k \in \mathbb{Z}_{++}$. Then $\exp(\text{adx}) = 1 + \text{adx} + \frac{(\text{adx})^2}{2!} + \dots + \frac{(\text{adx})^{k-1}}{(k-1)!}$.

CLAIM : $\exp(\text{adx}) \in \text{Aut}L$.

$$\begin{aligned} [\exp \delta(x), \exp \delta(y)] &= \left[\sum_{i=0}^{k-1} \frac{\delta^i x}{i!}, \sum_{j=0}^{k-1} \frac{\delta^j y}{j!} \right] \\ &= \sum_{n=0}^{2k-2} \left(\sum_{i=0}^n \left[\frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \right) \\ \text{pf.} \quad &= \sum_{n=0}^{2k-2} \frac{\delta^n [xy]}{n!} \quad \frac{\delta^n}{n!} [xy] = \sum_{i=0}^n \left[\frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \quad \text{where } \text{adx} \\ &= \sum_{n=0}^{k-1} \frac{\delta^n [xy]}{n!} \quad \delta^k = 0 \\ &= \exp \delta([xy]) \end{aligned}$$

is replaced by an arbitrary nilpotent derivation δ of L . Also given $\exp \delta = 1 + \eta$, its inverse is $1 - \eta + \eta^2 - \dots \pm \eta^{k-1}$ exists and therefore $\exp \delta$ is invertible. This gives the claim.

Leibniz rule

$$\frac{\delta^n}{n!}(xy) = \sum_{i=0}^n \frac{\delta^i x}{i!} \frac{\delta^{n-i} y}{(n-i)!}$$

pf. when $n = 0$, it trivially holds.

when $n > 0$, $\delta(\frac{\delta^{n-1}}{(n-1)!}(xy)) = \delta(\sum_{i=0}^{n-1} \frac{\delta^i x}{i!} \frac{\delta^{n-1-i} y}{(n-1-i)!}) = \sum_{i=0}^{n-1} \frac{\delta^{i+1} x}{i!} \frac{\delta^{n-1-i} y}{(n-1-i)!} + \sum_{i=0}^{n-1} \frac{\delta^i x}{i!} \frac{\delta^{n-i} y}{(n-1-i)!} = n \sum_{i=0}^n \frac{\delta^i x}{i!} \frac{\delta^{n-i} y}{(n-i)!}$ by the binomial identity. So the Leibniz rule follows by induction.

Definition 1.12. An automorphism of the form $\exp(\text{adx})$, adx nilpotent is called inner and the subgroup of $\text{Aut}L$ generated by these is denoted $\text{Int}L$, the set of inner automorphisms.

$\rightarrow$ it is a normal subgroup ($\phi \in \text{Aut}L$, $x \in L$, $\phi(\text{adx})\phi^{-1} = \text{ad}\phi(x) \rightarrow \phi \exp(\text{adx})\phi^{-1} = \exp(\text{ad}\phi(x))$).

CLAIM : If $L \subset \mathfrak{gl}(V)$ is an arbitrary linear Lie algebra, and $x \in L$ is nilpotent, then $(\exp x)y(\exp x)^{-1} = \exp \text{ad}(x)(y)$, $\forall y \in L$.

pf. $\text{adx} = \lambda_x + \rho_{-x}$ where $\begin{cases} \lambda_x & \text{left multiplication by } x \\ \rho_{-x} & \text{right multiplication by } -x \end{cases} \in \text{End}(\text{End}V)$. Then $\exp \text{adx} = \exp(\lambda_x + \rho_{-x}) = \exp \lambda_x \exp \rho_{-x} = \lambda_{\exp x} \cdot \rho_{\exp -x}$ and the stmt follows.

Definition 1.13. Define a sequence of ideals of L called the derived series by $L^{(0)} = L, L^{(1)} = [LL], L^{(2)} = [L^{(1)}L^{(1)}], \dots, L^{(i)} = [L^{(i-1)}L^{(i-1)}]$. Then L is said to be solvable if $L^{(n)} = 0$, $\exists n \in \mathbb{Z}_{++}$.

e.g. $\mathfrak{t}(n, F)$ is solvable.

pf. The basis for it consists of the matrix units e_{ij} , $i \leq j$ and the dimension is $\frac{n(n+1)}{2}$. Since $[e_{ii}e_{il}] = e_{il}$, $\forall i < l$, $\mathfrak{n}(n, F) \subset [LL]$ where it is the subalgebra of upper triangular nilpotent matrices. Since $\mathfrak{t}(n, F) = \mathfrak{d}(n, F) + \mathfrak{n}(n, F)$ and $\mathfrak{d}(n, F)$ is abelian, $\mathfrak{n}(n, F)$ is equal to the derived algebra of L . Define the level of e_{ij} by $j - i$. In the formula of commutators, assume $i < j, k < l$. If $i = l$, $[e_{ij}e_{kl}] = -e_{kj}$ with the level $j - k = j - i + l - k \geq 2$, two brackets of level $\geq 2^i$ must give level $\geq 2^{i-1}$.

So now consider the case where $i \neq l$. Then $[e_{ij}e_{kl}] = \begin{cases} e_{il} & j = k \\ 0 & \text{otherwise} \end{cases}$. Each e_{il} is a commutator of two matrices whose levels add up to that of e_{il} . Thus $L^{(2)}$ is spanned by e_{ij} of level ≥ 2 , $L^{(i)}$ by those of level $\geq 2^{i-1}$. Since $L^{(i)} = 0$ whenever $2^{i-1} > n - 1$ and therefore the stmt follows.

Proposition 1.2. Let L be a Lie algebra.

1. If L is solvable, then so are all subalgebras and homomorphic images of L .
2. If I is a solvable ideal of L s.t. L/I is solvable, then L itself is solvable.
3. If I, J are solvable ideals of L , then so is $I + J$.

pf.

- If K is a subalgebra of L , $K^{(i)} \subset L^{(i)}$ and therefore the subalgebras are also solvable. If $\phi : L \rightarrow M$ is an epimorphism, an easy induction on i shows $\phi(L^{(i)}) = M^{(i)}$ and therefore its homomorphic images are also solvable.
- Say $(L/I)^{(n)} = 0$. For the canonical homomorphism $\pi : L \rightarrow L/I$, 1 implies $\pi(L^{(n)}) = 0$, i.e. $L^{(n)} \subset I = \ker \pi$. By hypothesis, $\exists m \in \mathbb{Z}_{++}$ s.t. $I^{(m)} = 0 \rightarrow (L^{(n)})^{(m)} = L^{(n+m)} = 0$ so 2 follows.
- There exists an isomorphism between $(I+J)/J$ and $I/(I \cap J)$. As a homomorphic image of I , the RHS is solvable so $(I+J)/J$ is solvable. Then so is $I+J$ by 2.

$\rightarrow$ let L be an arbitrary Lie algebra and let S be a maximal solvable ideal. If I is any other solvable ideal of L , by the previous proposition 3, $S+I = S$ by maximality, hence $I \subset S$. Thus there exists a unique maximal solvable ideal of L , called the **radical** of L , denoted $\text{Rad}L$. If $\text{Rad}L = 0$, L is called **semisimple**.

1.2 Engel's Theorem

Definition 1.14. Define a sequence of ideals of L called the descending central series by $L^0 = L, L^1 = [LL], \dots, L^i = [LL^{i-1}]$. L is called **nilpotent** if $\exists n \in \mathbb{Z}_{++}$ s.t. $L^n = 0$.

$\rightarrow L^{(i)} \subset L^i, \forall i$ so nilpotent algebras are solvable but not vice versa.

pf. $L = \mathfrak{t}(n, \mathcal{F})$ is solvable. But $L^{(1)} = L^1 = \mathfrak{n}(n, \mathcal{F}) = [LL^1] = L^2 = \dots \forall i \geq 1$. Thus it is not nilpotent.

Proposition 1.3. Let L be a Lie algebra.

1. If L is nilpotent, then so are all subalgebras and homomorphic images of L .
2. If $L/Z(L)$ is nilpotent, then so is L .
3. If L is nilpotent, and nonzero, then $Z(L) \neq 0$.

pf.

- by the previous proposition, part 1
- $L^n \subset Z(L)$ implies $L^{n+1} = [LL^n] \subset [LZ(L)] = 0$
- for the descending central series, $\exists n \in \mathbb{Z}_{++}$ s.t. $L^{n-1} = [LL^{n-2}] \neq 0, L^n = [LL^{n-1}] = 0$. This means $0 \neq L^{n-1} \subset Z(L)$ and therefore $Z(L) \neq 0$.

$\rightarrow \exists n \in \mathbb{Z}_{++}$ s.t. $\text{adx}_1 \dots \text{adx}_n(y) = 0, \forall x_i, y \in L$ implies L to be nilpotent.

Definition 1.15. If L is any Lie algebra and $x \in L$, we call x *ad-nilpotent* if adx is a nilpotent endomorphism.

Lemma 1.1. Let $x \in \mathfrak{gl}(V)$ be a nilpotent endomorphism. Then adx is also nilpotent.

pf. Associate two endomorphisms $\begin{cases} \lambda_x & \text{left translation} \\ \rho_x & \text{right translation} \end{cases}$ to x . Then they are nilpotent since x is. Moreover $\lambda_x \rho_x(y) = xyx = \rho_x \lambda_x(y)$, i.e. they commute. In any ring the sum or difference of two commuting nilpotents is again nilpotent since x, y nilpotent with m, n s.t. $x^m = 0 = y^n$, $(x \pm y)^{m+n} = 0$. So $\text{adx} = \lambda_x - \rho_x$ is nilpotent.

Theorem 1.1. Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If L consists of nilpotent endomorphisms and $V \neq 0$, then there exists nonzero $v \in V$ for which $Lv = 0$.

pf.

- If $\dim L = 0$, the theorem is trivial.
- If $\dim L = 1$, a single nilpotent endomorphism has a nonzero vector in its kernel.
- If $\dim L > 1$, $\exists K \subsetneq L$ proper subalgebra of L . By the previous lemma, K acts via ad as a Lie algebra of nilpotent linear transformation on the vector space L , hence also on the vector space L/K . Because $\dim K < \dim L$, by the inductive hypothesis, $\exists x \in L$ with $x + K \neq K$ such that $x + K$ is killed by the image of K in $\mathfrak{gl}(L/K)$. Then $\forall y \in K$, $[yx] \in K$ while $x \notin K$, i.e. K is properly included in $N_L(K)$. Take K to be a maximal proper subalgebra of L . By the previous argument, $N_L(K) = L$, i.e. K is an ideal of L . If $\dim L/K$ were greater than 1, then the inverse image in L of one dimensional subalgebra of L/K would be a proper subalgebra properly containing K , a contradiction. Thus K has codimension 1. Thus $L = K + Fz$, $z \in L \setminus K$. By induction, $W = \{v \in V \mid Kv = 0\}$ is nonzero. Since K is an ideal, W is stable under L , i.e. $\forall x \in L$, $\forall y \in K$, $\forall w \in W$, $yxw = xyw - [xy]w = 0$. Choose $z \in L \setminus K$ as above, so the nilpotent endomorphism z has an eigenvector, i.e. $\exists v \in W$ for which $zv = 0$. Then $Lv = 0$ as desired.

Theorem 1.2 (Engel). If all elements of L are *ad-nilpotent*, then L is nilpotent.

pf. For any Lie algebra L all of whose elements are *ad-nilpotent*, the algebra $\text{ad}L \subset \mathfrak{gl}(L)$ satisfies the hypothesis of the previous theorem. Thus $\exists 0 \neq x \in L$ s.t. $[Lx] = 0$, i.e. $Z(L) \neq 0$. Now $L/Z(L)$ consists of *ad-nilpotent* elements and has smaller dimension than L . By inductive hypothesis, $L/Z(L)$ is nilpotent. The previous proposition, part 2, then implies that L itself is nilpotent.

Definition 1.16. If V is a finite dimensional vector space, a flag in V is a chain of subspaces $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V$, $\dim V_i = i$. If $x \in \text{End} V$ we say x stabilises this flag provided $xV_i \subset V_i$, $\forall i$.

Corollary 1.1. Under the hypotheses of the theorem, there exists a flag (V_i) in V stable under L with $xV_i \subset V_{i-1}$, $\forall i$.

pf. By Theorem 1.1, there exists $0 \neq v \in V$ s.t. $Lv = 0$. Set $V_1 = Fv$ and let $W = V/V_1$. Then the induced action of L on W is also by nilpotent endomorphisms. By induction on $\dim V$, W has a flag stabilised by L , whose inverse image in V does the trick.

Lemma 1.2. Let L be nilpotent, K an ideal of L . Then if $K \neq 0$, $K \cap Z(L) \neq 0$.

pf. L acts on K via the adjoint representation, so by Theorem 1.1, $\exists x \in K$ s.t. $[Lx] = 0$ so $x \in K \cap Z(L)$.

CLAIM : Let L be nilpotent, K a proper subalgebra of L . Then $N_L(K)$ includes K properly.

pf. By Lemma 1.1, K acts via ad as a Lie algebra of nilpotent linear transformations on the vector space L , hence also on the vector space L/K . By Theorem 1.1, $\exists x' \in L/K$ s.t. $Kx' = 0$. This implies $\forall y \in K$, $[yx] \in K$, $x \in L \setminus K$. Thus $x \in N_L(K) \setminus K$ and therefore it is properly contained.

CLAIM : Let L be nilpotent. Then L has an ideal of codimension 1.

pf. Let K be a maximal proper subalgebra of L . By the previous claim, $N_L(K) = L$. Then $[Ly] \subset K$, $\forall y \in K$ and therefore K is an ideal of L . If $\dim L/K$ were greater than 1, the inverse image in L of a one dimensional subalgebra of L/K would be a proper subalgebra properly containing K , a contradiction.

1.3 Theorems of Lie and Cartan

Assume throughout this subsection and the sequel that $\text{char} F = 0$.

Theorem 1.3. Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If $V \neq 0$, then V contains a common eigenvector for all the endomorphisms in L .

pf.

- $\dim L = 0$, it holds trivially
- $\dim L = 1$, its proper subalgebra $[L, L] = 0$ and L is abelian of dimension 1 and an eigenvector for a basis vector of L gives the theorem.

- $\dim L > 1$, since L is solvable, L properly includes $[LL]$. $L/[LL]$ being abelian, any subspace is automatically an ideal. Take a subspace of codimension 1, then its inverse image K is an ideal of codimension 1 in L . By the inductive hypothesis, $\exists v \in V$, $\exists \lambda : K \rightarrow F$ s.t. $xv = \lambda(x)v$, $\forall x \in K$. Fix this λ and set $W = \{w \in V | xw = \lambda(x)w, \forall x \in K\} \neq 0$.

– **CLAIM : L leaves W invariant.**

pf. $\forall w \in W$, $x \in L$, $\forall y \in K$, $yxw = xyw - [xy]w = \lambda(y)xw - \lambda([xy])w = \lambda(y)(xw)$ iff $\lambda([xy]) = 0$. So we only need to prove $\lambda([xy]) = 0$. Fix $w \in W$, $x \in L$. Let $n \in \mathbb{Z}_{++}$ be the smallest integer for which $w, xw, \dots, x^n w$ are linearly dependent. Let W_i be the subspace of V spanned by $w, xw, \dots, x^{i-1}w$, so $\dim W_n = n$, $W_n = W_{n+i}$, $\forall i \geq 0$ and x maps W_n into W_n . Relative to the basis $w, xw, \dots, x^{n-1}w$ of W_n ,

$$\begin{aligned} yx^i w &= yxx^{i-1}w = xyx^{i-1}w - [xy]x^{i-1}w \\ &= x\lambda(y)x^{i-1}w + xw' - [xy]x^{i-1}w \quad \text{by the inductive hypothesis,} \\ &\equiv \lambda(y)x^i w \pmod{W_i} \quad x \text{ maps } W_{i-1} \text{ into } W_i \end{aligned}$$

where $w' \in W_{i-1}$. This means $y \in K$ is represented by an upper triangular matrix whose diagonal entries are equal to $\lambda(y)$. Then $T|_{W_n}(y) = n\lambda(y)$. In particular, this is true for elements of K of the special form $[xy]$. But x, y both stabilise W_n , so $[xy]$ acts on W_n as the commutator of two endomorphisms of W_n and therefore its trace is 0. So $n\lambda([xy]) = 0$. Given $\text{char } F = 0$, $\lambda([xy]) = 0$.

Then write $L = K + Fz$. Since F is algebraically closed, $\exists v_0 \in W$ an eigenvector of z . Then v_0 is obviously a common eigenvector for L .

Corollary 1.2 (Lie). *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, $\dim V = n < \infty$. Then L stabilises some flag in V .*

pf. By the previous theorem with induction on $\dim V$.

Corollary 1.3. *Let L be solvable. Then there exists a chain of ideals of L , $0 = L_0 \subsetneq \dots \subsetneq L_n = L$ s.t. $\dim L_i = i$.*

Corollary 1.4. *Let L be solvable. Then $x \in [LL]$ implies $\text{ad}_L x$ is nilpotent. In particular, $[LL]$ is nilpotent.*

pf. Find a flag of ideals in the previous corollary. Relative to a basis $(x_1, \dots, x_n)$ of L for which $(x_1, \dots, x_i)$ spans L_i , the matrices of $\text{ad } L$ lie in $\mathfrak{t}(n, F)$. Thus the matrices of $[\text{ad } L, \text{ad } L] = \text{ad}_L[LL] \subset \mathfrak{n}(n, F)$ the derived algebra of $\mathfrak{t}(n, F)$. Thus $\text{ad}_L x$ is nilpotent for $x \in [LL]$. $\text{ad}_{[LL]} x$ is nilpotent, so $[LL]$ is nilpotent by Engel's theorem.

Definition 1.17. $x \in \text{End}V$ is semisimple iff the roots of its minimal polynomial over F are all distinct.

$\Leftrightarrow$ given F being algebraically closed, x is semisimple iff x is diagonalisable.

Proposition 1.4. Let V be a finite dimensional vector space over F , $x \in \text{End}V$.

1. There exists unique $x_s, x_n \in \text{End}V$ satisfying the conditions; $x = x_s + x_n$, $\begin{cases} x_s & \text{semisimple} \\ x_n & \text{nilpotent} \end{cases}$ which commute.
2. There exist polynomials $p(T), q(T)$ in one indeterminate, without constant term, s.t. $x_s = p(x)$, $x_n = q(x)$. In particular, x_s, x_n commute with any endomorphism commuting with x .
3. If $A \subset B \subset V$ are subspaces, and x maps B into A , then x_s, x_n also map B into A .

pf. Let $a_1, \dots, a_l$ be the distinct eigenvalues of x , so the characteristic polynomial is $\prod (T - a_i)^{m_i}$ where m_i denote the multiplicity. If $V_i = \ker(x - a_i \cdot 1)^{m_i}$, V is the direct sum of the subspaces $V_1, \dots, V_l$ each stable under x . On V_i , x clearly has characteristic polynomial $(T - a_i)^{m_i}$. By CRT, we get $p(T) \equiv \begin{cases} a_i \mod (T - a_i)^{m_i} \\ 0 \mod T \end{cases}$. Set $q(T) = T - p(T)$. Then $p(T), q(T)$ has zero constant term since $p(T) \equiv 0 \mod T$.

• 2,3

Set $\begin{cases} x_s = p(x) \\ x_n = q(x) \end{cases}$. Since they are polynomials in x , they commute with each other as well as all endomorphisms which commute with x .

They stabilise all subspaces of V stabilised by x , in particular the spaces V_i . The congruence $p(T) \equiv a_i \mod (T - a_i)^{m_i}$ shows the restriction of $x_s - a_i \cdot 1$ to V_i is zero for all i , hence x_s acts diagonally on V_i with single eigenvalue a_i . By definition, $x_n = x - x_s$ which makes it clear that x_n is nilpotent. Because $p(T), q(T)$ have no constant term, 3 follows.

• 1

Existence follows by the previous argument. If there exists $x = s + n$ another decomposition, $x_s - s = n - x_n$. Then by 2, all endomorphisms in the equation commute. Sums of commuting semisimple/nilpotent endomorphisms are semisimple/nilpotent, whereas only 0 can be both semisimple and nilpotent. Thus $x_s = s$, $x_n = n$.

→ the decomposition $x = x_s + x_n$ is called the **Jordan-Chevalley decomposition** of x .

Consider the adjoint representation of the Lie algebra $\mathfrak{gl}(V)$, V finite dimensional. If $x \in \mathfrak{gl}(V)$ is nilpotent, then so is adx by the lemma 1.1. Similarly, **if x is semisimple, then so is adx** .

pf. Choose a basis $(v_1, \dots, v_n)$ of V relative to which x has matrix $\text{diag}(a_1, \dots, a_n)$. Let $\{e_{ij}\}$ be the standard basis of $\mathfrak{gl}(V)$ relative to $(v_1, \dots, v_n)$. Then $e_{ij}(v_k) = \delta_{jk}v_i$. Then $\text{adx}(e_{ij}) = (a_i - a_j)e_{ij}$ so adx has diagonal matrix, relative to the chosen basis of $\mathfrak{gl}(V)$.

Lemma 1.3. *Let $x \in \text{End}V$, $x = x_s + x_n$ its Jordan decomposition. Then $\text{adx} = \text{adx}_s + \text{adx}_n$ is the Jordan decomposition of adx .*

pf. Since $\text{adx}_s, \text{adx}_n$ are semisimple and nilpotent respectively, they commute, since $[\text{adx}_s, \text{adx}_n] = \text{ad}[x_s, x_n] = 0$. Thus by the previous proposition 1, the lemma follows.

Lemma 1.4. *Let $\mathcal{U}$ be a finite dimensional F -algebra. Then $\text{Der}\mathcal{U}$ contains the semisimple and nilpotent parts in $\text{End}\mathcal{U}$ of all its elements.*

pf. If $\delta \in \text{Der}\mathcal{U}$, let $\sigma, \nu \in \text{End}\mathcal{U}$ be its semisimple and nilpotent parts respectively. If $a \in F$, set $\mathcal{U}_a = \{x \in \mathcal{U} | (\delta - a1)^k x = 0, \exists k\}$. Then $\mathcal{U}$ is the direct sum of those $\mathcal{U}_a$ for which a is an eigenvalue of δ and σ acts on $\mathcal{U}_a$ as scalar multiplication by a . For arbitrary $a, b \in F$, $\mathcal{U}_a \mathcal{U}_b \subset \mathcal{U}_{a+b}$ since $(\delta - (a+b)1)^n(xy) = \sum_{i=0}^n \binom{n}{i} ((\delta - a1)^{n-i}x) \cdot ((\delta - b1)^i y)$, $\forall x, y \in \mathcal{U}$. If $x \in \mathcal{U}_a$, $y \in \mathcal{U}_b$, $\sigma(xy) = (a+b)xy$ because $xy \in \mathcal{U}_{a+b}$. On the other hand, $\sigma(x)y + x\sigma(y) = (a+b)xy$. By directness of the sum $\mathcal{U} = \bigoplus \mathcal{U}_a$, $\sigma \in \text{Der}\mathcal{U}$. Then $\nu = \delta - \sigma \in \text{Der}\mathcal{U}$.

Lemma 1.5. *Let $A \subset B$ be two subspaces of $\mathfrak{gl}(V)$, $\dim V < \infty$. Set $M = \{x \in \mathfrak{gl}(V) | [xB] \subset A\}$. Suppose $x \in M$ satisfies $T(xy) = 0, \forall y \in M$. Then x is nilpotent.*

pf. Let $x = s + n$ be the Jordan decomposition of x . Fix a basis $v_1, \dots, v_m$ of V relative to which s has matrix $\text{diag}(a_1, \dots, a_m)$. Let E be the vector subspace of F over the prime field $\mathbb{Q}$ spanned by the eigenvalues $a_1, \dots, a_m$. Consider the dual space E^* , whose elements are the linear functions $f : E \rightarrow \mathbb{Q}$. Given such $f \in E^*$, let y be any element of $\mathfrak{gl}(V)$ whose matrix relative to the given basis is $\text{diag}(f(a_1), \dots, f(a_m))$. If $\{e_{ij}\}$ is the corresponding basis of $\mathfrak{gl}(V)$,

$$\begin{cases} \text{ads}(e_{ij}) = (a_i - a_j)e_{ij} \\ \text{ady}(e_{ij}) = (f(a_i) - f(a_j))e_{ij} \end{cases}.$$

Let $r(T) \in F[T]$ be a polynomial without constant term satisfying $r(a_i - a_j) = f(a_i) - f(a_j)$ for all pairs i, j . The existence of such $r(T)$ follows from Lagrange interpolation. Then $\text{ady} = r(\text{ads})$. Since ads is the semisimple part of adx by the lemma 1.3, it can be written as a polynomial in adx without constant term by the

proposition 1.4. Therefore $\text{ad}y = r(\text{ad}s)$ is also a polynomial in $\text{ad}x$ without constant term. By the hypothesis of the lemma, $\text{ad}x$ maps B into A , so $\text{ad}y(B) \subset A$ i.e. $y \in M$. By the hypothesis of the lemma, $T(xy) = 0 \rightarrow \sum a_i f(a_i) = 0$. The LHS is a $\mathbb{Q}$ -linear combination of elements of E . Applying f to the above equation gives $\sum (f(a_i))^2 = 0$. But the numbers $f(a_i)$ are rational, so this forces all of them to be 0, and therefore f must be 0 because the a_i span E . Then E^* is zero and its dual E is also 0 and the lemma follows.

Lemma 1.6. $T([xy]z) = T(x[yz])$ for all endomorphisms x, y, z of a finite-dimensional vector space.

pf. $\begin{cases} [xy]z = xyz - yxz \rightarrow T([xy]z) = T(xyz) - T(yxz) \\ x[yz] = xyz - xzy \rightarrow T(x[yz]) = T(xyz) - T(xzy) \end{cases}$ are equivalent given $T(ab) = T(ba)$.

Theorem 1.4 (Cartan's Criterion). *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. Suppose that $T(xy) = 0$, $\forall x \in [LL]$, $y \in L$. Then L is solvable.*

pf. Let $\begin{cases} A = [LL] \\ B = L \end{cases}$, so $M = \{x \in \mathfrak{gl}(V) \mid [xL] \subset [LL]\}$. Then $L \subset M$ trivially. By the hypothesis $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in L$. If $[xy]$ is a typical generator of $[LL]$, and if $z \in M$, the equation above shows $T([xy]z) = T(x[yz]) = T([yz]x)$. By the definition of M , $[yz] \in [LL]$ so $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in M$. Then by the previous lemma, every $x \in [LL]$ is nilpotent. Then by Engel's theorem, $[LL]$ is nilpotent and therefore L is solvable.

2 Semisimple Algebras

2.1 Killing Form

Definition 2.1. *Let L be any Lie algebra. If $x, y \in L$, define the Killing form $\kappa(x, y) = T(\text{ad}x \text{ad}y)$.*

$\rightarrow$ it is symmetric bilinear and associative in the sense that $\kappa([xy], z) = \kappa(x, [yz])$.

Lemma 2.1. *Let I be an ideal of L . If κ is the Killing form of L and κ_I be the Killing form of I , then $\kappa_I = \kappa|_{I \times I}$.*

pf. If W is a subspace of a finite dimensional vector space V , and ϕ an endomorphism of V mapping V into W , then $T\phi = T\phi|_W$, since we can extend a basis of W to a basis of V and calculate ϕ . Then if $x, y \in I$, $\text{ad}x \text{ad}y$ is an endomorphism of L , mapping L into I , so its trace $\kappa(x, y)$ coincides with the trace $\kappa_I(x, y)$ of $\text{ad}x \text{ad}y|_I = \text{ad}_I x \text{ad}_I y$.

Definition 2.2. A symmetric bilinear form $\beta(x, y)$ is called nondegenerate if its radical S is 0, where $S = \{x \in L \mid \beta(x, y) = 0, \forall y \in L\}$.

$\rightarrow S$ is an ideal of L , given the associativity of the Killing form.

Theorem 2.1. Let L be a Lie algebra. Then L is semisimple iff its Killing form is nondegenerate.

pf.

- Assume L is semisimple. Then $\text{Rad}L = 0$. Let S be the radical of κ . By definition, $T(\text{adx} \text{ ad} y) = 0, \forall x \in S, \forall y \in L$. By the Cartan criterion, $\text{ad}_L S$ is solvable, hence S is solvable. Since S is an ideal of L , $S \subset \text{Rad}L = 0$ and κ is nondegenerate.
- Let $S = 0$ and I be an abelian ideal of L . Suppose $x \in I, y \in L$. Then $\text{adx} \text{ ad} y$ maps $L \rightarrow L \rightarrow I$ and $(\text{adx} \text{ ad} y)^2$ maps L into $[II] = 0$. This means $\text{adx} \text{ ad} y$ is nilpotent, hence $0 = T(\text{adx} \text{ ad} y) = \kappa(x, y)$ so $I \subset S = 0$.

$\rightarrow S \subset \text{Rad}L$ but not vice versa.

pf. Let L be a Lie algebra with basis x, y, z s.t.
$$\begin{cases} [xy] = z \\ [xz] = -y \\ [yz] = 0 \end{cases} \text{ . Then } \text{adx} =$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \text{ady} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \text{ad}(z) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and therefore its } \kappa = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

So $\text{Rad}L = L \not\subset \text{span}(y, z) = S \equiv \text{radical of } \kappa$.

Definition 2.3. A Lie algebra L is said to be the direct sum of ideals $I_1, \dots, I_t$ provided $L = I_1 + \dots + I_t$, denoted $L = I_1 \oplus \dots \oplus I_t$.

$\rightarrow [I_i I_j] \subset I_i \cap I_j = 0, \forall i \neq j$.

Theorem 2.2. Let L be semisimple. Then there exist ideals $L_1, \dots, L_t$ of L which are simple s.t. $L = L_1 \oplus \dots \oplus L_t$. Every simple ideal of L coincides with one of the L_i . Moreover, the Killing form of L_i is the restriction of κ to $L_i \times L_i$.

pf.

- **CLAIM : Let I be an arbitrary ideal of L . Then $L = I \oplus I^\perp$.**

pf. $I^\perp = \{x \in L : \kappa(x, y) = 0, \forall y \in I\}$ is also an ideal by the associativity of κ . By Cartan's criterion applied to $I \cap I^\perp$, it is solvable, hence 0 as an ideal of the semisimple L . Thus since $\dim I + \dim I^\perp = \dim L$, $L = I \oplus I^\perp$.

- Existence

If L has no proper nonzero ideal, then L itself is simple and there is nothing to prove. Otherwise, let L_1 be a minimal nonzero ideal. Then by the first claim, $L = L_1 \oplus L_1^\perp$. In particular, any ideal of L_1 is also an ideal of L and therefore L_1 is semisimple, and by minimality it is simple. For the same reason $L_1^\perp$ is semisimple. By inductive hypothesis, it splits into a direct sum of simple ideals, which are also ideals of L .

- Uniqueness

If I is any simple ideal of L , then $[IL]$ is also an ideal of L , nonzero because $Z(L) = 0$. Then $[IL] = I$ by simplicity. On the other hand, $[IL] = [IL_1] + \cdots + [IL_t]$ so all but one summand must be 0. Say $[IL_i] = I$. Then $I \subset L_i$, which forces $I = L_i$ by the simplicity of L_i .

Corollary 2.1. *If L is semisimple, $L = [LL]$ and all ideals and homomorphic images of L are semisimple. Moreover, each ideal of L is a sum of certain simple ideals of L .*

For any Lie algebra L , $\text{ad}L$ is an ideal in $\text{Der}L$.

pf. $\forall y, [\delta, \text{ad}x](y) = \delta \text{ad}x(y) - \text{ad}x\delta(y) = \delta([xy]) - [x, \delta y] = [\delta x, y] + [x, \delta y] - [x, \delta y] = [\delta x, y] = \text{ad}(\delta x)(y) \Leftrightarrow [\delta, \text{ad}x] = \text{ad}(\delta x), \forall x \in L, \forall \delta \in \text{Der}L$. So the stmt follows.

Theorem 2.3. *If L is semisimple, then $\text{ad}L = \text{Der}L$.*

pf. Since L is semisimple, $Z(L) = 0$. Thus $L \rightarrow \text{ad}L$ is an isomorphism of Lie algebras. In particular, $M = \text{ad}L$ has nondegenerate Killing form by the theorem 2.1. If $D = \text{Der}L$, $[DM] \subset M$. Then by the lemma 2.1, κ_M is the restriction to $M \times M$ of the Killing form κ_D of D . In particular, if $I = M^\perp$ is the subspace of D orthogonal to M under κ_D , then the nondegeneracy of κ_M forces $I \cap M = 0$. Both I, M are ideals of D so $[IM] = 0$. If $\delta \in I$, $\text{ad}\delta x = 0, \forall x \in L$ by the previous stmt, so $\delta x = 0, \forall x \in L$ because ad is one to one and $\delta = 0$. So $I = 0, \text{Der}L = M = \text{ad}L$.

→ For any semisimple Lie algebra L , if $\mathcal{U}$ is any F -algebra of finite dimension, $\text{Der}\mathcal{U}$ contains the semisimple and nilpotent parts in $\text{End}\mathcal{U}$ of all its elements. In particular, since $\text{Der}L$ coincides with $\text{ad}L$ while $L \rightarrow \text{ad}L$ is one to one, each $x \in L$ determines unique elements $s, n \in L$ s.t. $\text{ad}x = \text{ad}s + \text{ad}n$ is the Jordan decomposition of $\text{ad}x$. This means $x = s + n, [sn] = 0, \begin{cases} s & \text{ad-semisimple} \\ n & \text{ad-nilpotent} \end{cases}$ and

$\begin{cases} s = x_s & \text{the semisimple part} \\ n = x_n & \text{the nilpotent part} \end{cases}$ of x .

2.2 Complete Reducibility of Representations

Definition 2.4. Let L be a Lie algebra. A vector space V , endowed with an operation $L \times V \rightarrow V$ $(x, v) \mapsto xv$ is called an L -module if the following conditions are satisfied :

$$\begin{cases} (ax + by)v = a(xv) + b(yv), & x(av + bw) = a(xv) + b(xw) \\ [xy]v = xyv - yxv \end{cases}, \quad \forall x, y \in L. \quad \forall v, w \in V, \quad a, b \in F.$$

e.g. If $\phi : L \rightarrow \mathfrak{gl}(V)$ is a representation of L , then V may be viewed as an L -module via the action $xv = \phi(x)(v)$. Conversely, given an L -module V , this equation defines a representation $\phi : L \rightarrow \mathfrak{gl}(V)$.

Definition 2.5. A homomorphism of L -modules is a linear map $\phi : V \rightarrow W$ s.t. $\phi(xv) = x\phi(v)$. When ϕ is an isomorphism of vector spaces, we call it an isomorphism of L -modules.

Definition 2.6. An L -module V is called irreducible if it has precisely two L -submodules- itself and 0.

Definition 2.7. V is called completely reducible if V is a direct sum of irreducible L -submodules or equivalently if each L -submodule W of V has a complement W' .

→ if it is a direct sum of irreducible L -submodules, its any submodule W is also a direct sum of irreducible L -submodules. Then the summand that does not belong to its submodule also makes an L -submodule $W^\perp$ s.t. $W \cap W^\perp = 0$, $W + W^\perp = V$ so $V = W \oplus W^\perp$. Conversely, if any submodule of V has its complement, pick a minimal submodule of V . If there is no proper submodule, V is irreducible and there is nothing to prove. Otherwise, pick a minimal proper submodule V_1 of V . Then by minimality, it is irreducible. Then it has complement $V_1^\perp$ with $\dim V_1^\perp < \dim V$. Then by the inductive hypothesis it is a direct sum of irreducible L -submodules and therefore $V = V_1 \oplus V_1^\perp$ is.

Given a representation $\phi : L \rightarrow \mathfrak{gl}(V)$, the associative algebra generated by $\phi(L)$ in $\text{End}V$ leaves invariant precisely the same subspaces as L . Therefore, all the usual results for modules over associative rings hold for L as well.

SCHUR'S LEMMA : Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be irreducible. Then the only endomorphisms of V commuting with all $\phi(x)$ are the scalars.

L itself is an L -module, and then an L -submodule is just an ideal, so a simple algebra L is irreducible as L -module, while a semisimple algebra is completely reducible.

MAKING NEW MODULES out of OLD ONES

- dual module

Let V be an L -module. Then the dual vector space V^* becomes an L -module with $(xf)(v) = -f(xv)$, $\forall f \in V^*$. Then $([xy]f)(v) = -f([xy]v) = -f(xyv - yxv) = -f(xyv) + f(yxv) = (xf)(yv) - (yf)(xv) = -(yxf)(v) + (xyf)(v) = ((xy - yx)f)(v)$.

- tensor-product

If V, W are L -modules, let $V \otimes W$ be the tensor product over F of the underlying vector spaces. For Lie algebras, define $x(v \otimes w) = xv \otimes w + v \otimes xw$. Then $[xy](v \otimes w) = [xy]v \otimes w + v \otimes [xy]w = (xyv - yxv) \otimes w + v \otimes (xyw - yxw) = (xyv \otimes w + v \otimes xyw) - (yxv \otimes w + v \otimes yxw) = (xy - yx)(v \otimes w)$.

$\rightarrow$ given a vector space V over F , $\exists$ an isomorphism $V^* \otimes V \xrightarrow{f \otimes v} \text{End} V$ ($w \mapsto f(w)v$) enables us to consider $\text{End} V$ as an L -module via the isomorphism. The action of L on $\text{End} V$ is $(x \cdot \psi)(v) = x\psi(v) - \psi(xv)$, $\forall \psi \in \text{End} V$, $\forall v \in V$.

Let L be semisimple and $\phi : L \rightarrow \mathfrak{gl}(V)$ be a faithful representation of L . Define a symmetric bilinear form $\beta(x, y) = T(\phi(x)\phi(y))$ on L . The form β is associative, so its radical S is an ideal of L . Moreover $\phi(S) \cong S$ is solvable, so $S = 0$ and β is non-degenerate. Let L be semisimple, β any nondegenerate symmetric associative bilinear form on L . If $(x_1, \dots, x_n)$ is a basis of L , there is a uniquely determined dual basis $(y_1, \dots, y_n)$ relative to β satisfying $\beta(x_i, y_j) = \delta_{ij}$. If $x \in L$,

$$\begin{aligned} a_{ik} &= \sum_j a_{ij} \beta(x_j, y_k) = \beta([xx_i], y_k) && \text{by the definition of } \beta \\ &= \beta(-[x_i x], y_k) && \text{by the skew-symmetry of bracket} \\ \text{Then} \quad &= \beta(x_i, -[xy_k]) && \text{by the associativity of } \beta \\ &= -\sum_j b_{kj} \beta(x_i, y_j) && = -b_{ki} \end{aligned}$$

$L \rightarrow \mathfrak{gl}(V)$ is any representation of L , write $c_\phi(\beta) = \sum_i \phi(x_i)\phi(y_i) \in \text{End} V$ where

$$\begin{aligned} [\phi(x)c_\phi(\beta)] &= \sum_i [\phi(x)\phi(x_i)]\phi(y_i) + \sum_i \phi(x_i)[\phi(x)\phi(y_i)] \\ &= \sum_{ij} a_{ij} \phi(x_j)\phi(y_i) + \sum_{ij} b_{ij} \phi(x_i)\phi(y_j) \\ &= 0 \end{aligned}$$

x_i, y_i run over dual bases relative to β . Then

Thus $c_\phi(\beta)$ is an endomorphism of V commuting with $\phi(L)$.

$\rightarrow c_\phi(\beta)$ is called the Casimir element of ϕ and its trace is $\sum_i T(\phi(x_i)\phi(y_i)) = \sum_i \beta(x_i, y_i) = \dim L$. If ϕ is also irreducible, by Schur's lemma, it should be scalar.

e.g. $L = \mathfrak{sl}(2, F)$, $V = F^2$, ϕ is the identity map $L \rightarrow \mathfrak{gl}(V)$. Let (x, h, y) be the standard basis of L . Then its dual basis relative to the trace form is $(y, h/2, x)$ so

$$c_\phi = xy + \frac{h^2}{2} + yx = \begin{bmatrix} 3/2 & 0 \\ 0 & 3/2 \end{bmatrix}.$$

When ϕ is not faithful, we can factor out $\ker \phi$ and get L' the sum of the remaining simple ideals after factoring out $\ker \phi$. Then $\phi|_{L'}$ is faithful and we can get the Casimir element for L' . Then $\phi(L) = \phi(L')$ commute with the Casimir element.

Lemma 2.2. *Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a representation of a semisimple Lie algebra L . Then $\phi(L) \subset \mathfrak{sl}(V)$. In particular, L acts trivially on any one dimensional L -module.*

pf. $L = [L, L]$ and $\mathfrak{sl}(V)$ is the derived algebra of $\mathfrak{gl}(V)$, the lemma follows.

Theorem 2.4 (Weyl). *Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a finite dimensional representation of a semisimple Lie algebra. Then ϕ is completely reducible.*

pf.

- V has an L -submodule W of codimension 1

Since by the previous lemma, L acts trivially on V/W , wlog we may denote this module F and get a s.e.s. $0 \rightarrow W \rightarrow V \rightarrow F \rightarrow 0$.

– **CLAIM : wlog we can assume W is an irreducible L -module.**

Let W' be a proper nonzero submodule of W . Then we get another s.e.s. $0 \rightarrow W/W' \rightarrow V/W' \rightarrow F \rightarrow 0$. By the inductive hypothesis, it splits, so there is a one dimensional L -submodule of V/W' complementary to W/W' . Let it be $\tilde{W}/W'$. Then there exists another s.e.s $0 \rightarrow W' \rightarrow \tilde{W} \rightarrow F \rightarrow 0$ with $\dim W' < \dim W$, so by the inductive hypothesis, there exists a submodule X complementary to W' , i.e. $\tilde{W} = W' \oplus X$. But $V/W' = W/W' \oplus \tilde{W}/W'$ so $V = W \oplus X$, given $W \cap X = 0$ and dimension matching.

Let $c = c_\phi$ be the Casimir element of ϕ . Since c commutes with $\phi(L)$, c is actually an L -module endomorphism of V . In particular, $c(W) \subset W$ and $\ker c$ is an L -submodule of V . Because L acts trivially on V/W , c must do likewise as a linear combination of products of elements $\phi(x)$. So c has trace 0 on V/W . On the other hand, c acts as a scalar on the irreducible L -submodule W by Schur's lemma. This cannot be 0, because that would force $T|_V(c) = 0$, a contradiction. Thus $\ker c$ is a one-dimensional L -submodule of V which intersects W trivially, which is the desired complement of W .

- General case

Let W be a nonzero submodule of V . Then there exists a s.e.s. $0 \rightarrow W \rightarrow$

$V \rightarrow V/W \rightarrow 0$. Let $\text{hom}(V, W)$ be the space of linear maps $V \rightarrow W$, viewed as L -module. Let $\mathcal{V}$ be the subspace of $\text{hom}(V, W)$ consisting of those maps whose restriction to W is a scalar multiplication. Then $f|_W = a1_W$, $\forall x \in L$, $\forall w \in W$, $(xf)(w) = xf(w) - f(xw) = a(xw) - a(xw) = 0 \rightarrow (xf)|_W = 0$ so $\mathcal{V}$ is an L -submodule. Let $\mathcal{W}$ be the subspace of $\mathcal{V}$ consisting of those f whose restriction to W is zero. Then $\mathcal{W}$ is also an L -submodule and L maps $\mathcal{V}$ to $\mathcal{W}$. Moreover $\mathcal{V}/\mathcal{W}$ has dimension one because $f \in \mathcal{V}$ is determined by the scalar $f|_W \bmod \mathcal{W}$. This gives the first situation. Then $\mathcal{V}$ has a one-dimensional submodule complementary to $\mathcal{W}$. Let $f : V \rightarrow W$ span it, so after multiplying by a nonzero scalar, we may assume $f|_W = 1_W$. Since f maps V into W and acts as 1_W on W , we can say $V = W \oplus \ker f$.

Theorem 2.5. *Let $L \subset \mathfrak{gl}(V)$ be a semisimple linear Lie algebra with V finite dimensional. Then L contains the semisimple and nilpotent parts in $\mathfrak{gl}(V)$ of all its elements. In particular, the abstract and usual Jordan decompositions in L coincide.*

pf. Let $x \in L$ arbitrary, with Jordan decomposition $x = x_s + x_n$ in $\mathfrak{gl}(V)$. Since $\text{ad}x(L) \subset L$, $\begin{cases} \text{ad}x_s(L) \subset L \\ \text{ad}x_n(L) \subset L \end{cases}$ where $\text{ad} = \text{ad}_{\mathfrak{gl}(V)}$, i.e. $x_s, x_n \in N_{\mathfrak{gl}(V)}(L) = N$, which is a Lie subalgebra of $\mathfrak{gl}(V)$ including L as an ideal. If W is any L -submodule of V , define $L_W = \{y \in \mathfrak{gl}(V) | y(W) \subset W, T(y|_W) = 0\}$. For example, $L_V = \mathfrak{sl}(V)$. Since $L = [LL]$, L lies in all such L_W . Set $L' =$ intersection of N with all spaces L_W . Then L' is a subalgebra of N including L as an ideal. Then if $x \in L$, x_s, x_n also lie in L_W and therefore in L' .

• **CLAIM :** $L = L'$.

pf. L' being finite dimensional L -module, by Weyl's theorem $L' = L + M$ for some L -submodule M , where the sum is direct. But $[LL'] \subset L$ given $L' \subset N$ so the action of L on M is trivial. Let W be any irreducible L -submodule of V . If $y \in M$, $[Ly] = 0$ so by Schur's lemma, y acts on W as a scalar. On the other hand, $T(y|_W) = 0$ because $y \in L_W$. Therefore y acts on W as zero. V can be written as a direct sum of irreducible L -submodules by Weyl's theorem so $y = 0$. Thus $M = 0$, $L = L'$.

Corollary 2.2. *Let L be a semisimple Lie algebra, $\phi : L \rightarrow \mathfrak{gl}(V)$ a finite dimensional representation of L . If $x = s + n$ is the abstract Jordan decomposition of $x \in L$, $\phi(x) = \phi(s) + \phi(n)$ is the usual Jordan decomposition of $\phi(x)$.*

pf. The algebra $\phi(L)$ is spanned by the eigenvectors of $\text{ad}_{\phi(L)}\phi(s)$ since L has this property relative to ads , and therefore $\text{ad}_{\phi(L)}\phi(s)$ is semisimple. Similarly, $\text{ad}_{\phi(L)}\phi(n)$ is nilpotent and it commutes with $\text{ad}_{\phi(L)}\phi(s)$. Thus $\phi(x) = \phi(s) + \phi(n)$

is the abstract Jordan decomposition of $\phi(x)$ in the semisimple Lie algebra $\phi(L)$. Then by the previous theorem, the corollary follows.

Definition 2.8. A Lie algebra L for which $\text{Rad}L = Z(L)$ is called *reductive*.

e.g. $\begin{cases} L \text{ abelian} \\ L = \mathfrak{gl}(n, F) \end{cases}$, then it is reductive.

- **If L is reductive, L is completely reducible $\text{ad}L$ -module.**

pf. By the hypothesis $L/Z(L)$ is semisimple. Since $\text{ad}L$ acts trivially on $Z(L)$, the adjoint representation of L on L factors through $L/Z(L)$. Thus L is an $L/Z(L)$ -module. $\phi : L/Z(L) \rightarrow \mathfrak{gl}(V)$ is a representation of a semisimple Lie algebra. Then ϕ is completely reducible by Weyl's theorem. Thus L is completely reducible.

- **If L is a completely reducible $\text{ad}L$ -module, then L is reductive.**

pf. There is an adjoint representation $\phi : L \rightarrow \mathfrak{gl}(L)$, let $R = \text{Rad}L$ be its radical. Then since it is an ideal of L , it is an $\text{ad}L$ -submodule of L . Then, $L = R \oplus S$ by L being completely reducible. Because $\text{ad}L$ -submodules of the adjoint module are ideals, S is also an ideal. Thus $\begin{cases} [RS] \subseteq R \\ [RS] \subseteq S \end{cases}$. Thus $[RS] \subseteq R \cap S = 0 \rightarrow [RS] = 0$. Since R is solvable, $[RR] \subsetneq R$ unless $R = 0$ and $[RR]$ is also an $\text{ad}L$ -submodule. By complete reducibility, choose an $\text{ad}L$ -submodule $T \subseteq R$ s.t. $R = [RR] \oplus T$. Given T an ideal, $\begin{cases} [RT] \subseteq T \\ [RT] \subseteq [RR] \end{cases} \rightarrow [RT] = 0, R = [RR] \oplus T$. Thus $[RR] = [R[RR]]$. If $[RR] \neq 0$, $R^{(1)} = R^{(2)} = \dots$, a contradiction. Thus $[RR] = 0$ and $R \subseteq Z(L)$. Since $Z(L) \subseteq R$ the stmt follows.

- **If L is reductive, then all finite dimensional representations of L in which $Z(L)$ is represented by semisimple endomorphisms are completely reducible.**

pf. By the first stmt, it is a completely reducible $\text{ad}L$ -module and $L = Z(L) \oplus [LL]$ where $[LL]$ is semisimple. Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be finite dimensional with every element of $Z(L)$ represented semisimply. By Weyl's theorem, it is completely reducible as $[LL]$ -module. $[Z(L), L] = 0$ implies $[\phi(z)\phi(x)] = 0, \forall z \in Z(L), \forall x \in [LL]$. Since $\phi(z)$ are commuting semisimple endomorphisms over an algebraically closed field they are simultaneously diagonalisable. Thus $V = \bigoplus_{\lambda} V_{\lambda}$ where $\forall V_{\lambda}$ is a common eigenspace for $Z(L)$. Since $\phi(z)$ commute with $\phi([LL])$, $xV_{\lambda} \subseteq V_{\lambda}, \forall x \in [LL]$. So V_{λ} is an $[LL]$ -module

and by Weyl's theorem $V_\lambda = \bigoplus_j V_{\lambda,j}$ where $V_{\lambda,j}$ are irreducible $[LL]$ -module. But $Z(L)$ acts on all $V_{\lambda,j}$ by the scalar $\lambda(z)$, $V_{\lambda,j}$ is actually an L -submodule. If it is a proper L -submodule, then it is an $[LL]$ -submodule, contradicting irreducibility. Hence $V = \bigoplus_{\lambda,j} V_{\lambda,j}$ is a decomposition into irreducible L -modules.

2.3 Representations of $\mathfrak{sl}(2, F)$

Let L denote $\mathfrak{sl}(2, F)$ whose standard basis consists of $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then $[hx] = 2x$, $[hy] = -2y$, $[xy] = h$.

Let V be an arbitrary L -module. Since h is semisimple, h acts diagonally on V by the previous corollary. Then V is decomposed as direct sum of eigenspaces $V_\lambda = \{v \in V | hv = \lambda v\}$, $\lambda \in F$. Whenever $V_\lambda \neq 0$, λ is called a weight of h in V and we call V_λ a weight space.

Lemma 2.3. If $v \in V_\lambda$, $\begin{cases} xv \in V_{\lambda+2} \\ yv \in V_{\lambda-2} \end{cases}$.

pf. $\begin{cases} h(xv) = [hx]v + xhv = 2xv + \lambda xv = (\lambda + 2)xv \\ h(yv) = [hy]v + yhv = -2yv + \lambda yv = (\lambda - 2)yv \end{cases}$
 $\rightarrow x, y$ are represented by nilpotent endomorphisms of V .

Since $\dim V < \infty$ and $V = \bigoplus_{\lambda \in F} V_\lambda$ is direct, there must exist $V_\lambda \neq 0$ s.t. $V_{\lambda+2} = 0$. For such λ , any nonzero vector in V_λ will be called a maximal vector of weight λ .

Assume V is an irreducible L -module. Choose a maximal vector, say $v_0 \in V$ and set $v_{-1} = 0$, $v_i = \frac{1}{i!} y^i v_0$, $\forall i \geq 0$.

Lemma 2.4. $\begin{cases} (i) & hv_i = (\lambda - 2i)v_i \\ (ii) & yv_i = (i + 1)v_{i+1} \\ (iii) & xv_i = (\lambda - i + 1)v_{i-1} \end{cases}$

pf.

- for $i = 0$, $hv_0 = \lambda v_0$
 $i > 0$, $hv_i = \frac{1}{i} h y v_{i-1} = \frac{1}{i} ([hy]v_{i-1} + y h v_{i-1}) = \frac{1}{i} (-2y v_{i-1} + (\lambda - 2(i-1))y v_{i-1}) = \frac{1}{i} ((\lambda - 2i)y v_{i-1}) = (\lambda - 2i)v_i$
- $(i + 1)v_{i+1} = (i + 1) \frac{1}{(i+1)!} y^{i+1} v_0 = y v_i$

$$\begin{aligned} i > 0 \rightarrow &= (\lambda - 2(i - 1))v_{i-1} + (\lambda - i + 2)yv_{i-2} && \text{i and inductive hypothesis} \\ &= (\lambda - 2i + 2)v_{i-1} + (i - 1)(\lambda - i + 2)v_{i-1} && \text{by (ii)} \\ &= i(\lambda - i + 1)v_{i-1} \end{aligned}$$

→ (i) implies the nonzero v_i are all linearly independent. But $\dim V < \infty$. Let m be the smallest integer for which $\begin{cases} v_m \neq 0 \\ v_{m+1} = 0 \end{cases}$. Then the subspace of V with basis $(v_0, \dots, v_m)$ is an L -submodule different from 0. Because V is irreducible, this subspace must be all of V and moreover relative to the ordered basis $(v_0, \dots, v_m)$, the matrices of the endomorphisms representing x, y, h can be written down explicitly. Also for $i = m + 1$, $xv_i = 0$ in the LHS of (iii), whereas the RHS is $(\lambda - m)v_m$. Since $v_m \neq 0$, $\lambda = m$, i.e. the weight of a maximal vector is a nonnegative integer. We call it the highest weight of V . Moreover each weight μ occurs with multiplicity one by (i). In particular, since V determines λ uniquely, the maximal vector v_0 is the only possible one in V .

Theorem 2.6. *Let V be an irreducible module for $L = \mathfrak{sl}(2, F)$.*

1. *Relative to h , V is the direct sum of weight spaces V_μ , $\mu = m, m-2, \dots, -(m-2), -m$ where $m+1 = \dim V$ and $\dim V_\mu = 1$ for each such μ .*
2. *V has up to nonzero scalar multiples a unique maximal vector, whose weight is m .*
3. *The action of L on V is given explicitly by the above formulas, if the basis is chosen in the prescribed fashion. In particular, there exists at most one irreducible L -module up to isomorphism of each possible dimension $m+1$, $m \geq 0$.*

pf. By the preceding paragraph.

Corollary 2.3. *Let V be any finite dimensional L -module, $L = \mathfrak{sl}(2, F)$. Then the eigenvalues of h on V are all integers and each occurs along with its negative. Moreover, in any decomposition of V into direct sum of irreducible submodules, the number of summands is precisely $\dim V_0 + \dim V_1$.*

pf. If $V = 0$ then nothing to prove. If $V \neq 0$, by Weyl's theorem, we can write V as direct sum of irreducible submodules. Then by the previous theorem, the first part of the corollary follows. Each irreducible L -module has a unique occurrence of either the weight 0 or else the weight 1, but not both so the last part of the corollary follows.

Existence of the maximal vector

pf. Look at the subalgebra B spanned by h, x . It is a solvable subalgebra of $\mathfrak{gl}(V)$ with $\dim V < \infty$. Since $V \neq 0$, V contains a common eigenvector v for all the endomorphisms in B by Lie's theorem ($[BB] = Fx$, $[FxFx] = 0$).

• **CLAIM : v is a maximal vector.**

pf. Since v is a common eigenvector for B , $\exists \lambda, \mu \in F$ s.t. $\begin{cases} hv = \lambda v \\ xv = \mu v \end{cases}$. Then $2xv = [hx]v = hxv - xhv = h(\mu v) - x(\lambda v) = \mu\lambda v - \lambda\mu v = 0$. Since $\text{char} F = 0$, $xv = 0$ so v is a maximal vector.

2.4 Root Space Decomposition

Let L denote a semisimple Lie algebra. If L consisted entirely of nilpotent elements, then L would be nilpotent by Engel's theorem. This not being the case, we can find $x \in L$ whose semisimple part x_s in the abstract Jordan decomposition is nonzero. Then L possesses nonzero subalgebras consisting of semisimple elements. Call such a subalgebra **toral**.

Lemma 2.5. *A toral subalgebra of L is abelian.*

pf. Let T be toral. Since adx , $x \in T$ is diagonalisable since adx is semisimple and F is algebraically closed, then if $\text{ad}_T x$ has no nonzero eigenvalues, then $\text{ad}_T x = 0$, $\forall x \in T$. Suppose the contrary. Then $\exists 0 \neq y \in T$ s.t. $[xy] = ay$, $a \neq 0$. Then $\text{ad}_T y(x) = -ay$ shows that $\text{ad}_T y(x)$ is an eigenvector of $\text{ad}_T y$ of eigenvalue 0 ($\text{ad}_T y(\text{ad}_T y(x)) = \text{ad}_T y(-ay) = 0$). On the other hand, we can write x as a linear combination of eigenvectors of $\text{ad}_T y$ since y is semisimple. Thus after applying $\text{ad}_T y$ to x , all that is left is a combination of eigenvectors which belong to nonzero eigenvalues, if any, a contradiction. Thus the lemma follows.

e.g. $L = \mathfrak{sl}(2, F)$. Let H be a maximal toral subalgebra. Then $x, y \notin H$ since they are nilpotent. But $Fh \subset H$. This forces $Fh = H$.

Fix a maximal toral subalgebra H of L , i.e. a toral subalgebra not properly included in any other. Since H is abelian by the previous lemma, $\text{ad}_L H$ is a commuting family of semisimple endomorphisms of L . Then $\text{ad}_L H$ is simultaneously diagonalisable, i.e. L is the direct sum of the subspaces $L_\alpha = \{x \in L \mid [hx] = \alpha(h)x, \forall h \in H\}$, where α ranges over H^* . Here $L_0 = C_L(H)$, the centraliser of H . Then $H \subseteq L_0$ by the lemma. The set of all nonzero $\alpha \in H^*$ for which $L_\alpha \neq 0$ is denoted by Φ .

Definition 2.9. The elements of Φ are called the roots of L relative to H and a root space decomposition is

$$L = C_L(H) \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$$

Proposition 2.1. $\forall \alpha, \beta \in H^*$, $[L_\alpha L_\beta] \subset L_{\alpha+\beta}$. If $x \in L_\alpha$, $\alpha \neq 0$, then adx is nilpotent. If $\alpha, \beta \in H^*$ and $\alpha + \beta \neq 0$, then L_α is orthogonal to L_β relative to the Killing form κ of L .

pf.

- $\forall \begin{cases} x \in L_\alpha \\ y \in L_\beta \end{cases}, h \in H$ implies

$$\begin{aligned} \text{adh}([xy]) &= [h[xy]] = -[x[yh]] - [y[hx]] && \text{by the Jacobi identity} \\ &= [[hx]y] + [x[hy]] && \text{by the skew-symmetry of bracket} \\ &= \alpha(h)[xy] + \beta(h)[xy] \\ &= (\alpha + \beta)(h)[xy] \end{aligned}$$
- find $h \in H$ for which $(\alpha + \beta)(h) \neq 0$. Then if $x \in L_\alpha$, $y \in L_\beta$, $\kappa([hx], y) = -\kappa([xh], y) = -\kappa(x, [hy]) \Leftrightarrow \alpha(h)\kappa(x, y) = -\beta(h)\kappa(x, y) \Leftrightarrow (\alpha + \beta)(h)\kappa(x, y) = 0$, which forces $\kappa(x, y) = 0$.

Corollary 2.4. The restriction of the Killing form to $L_0 = C_L(H)$ is nondegenerate.

pf. κ is nondegenerate. But L_0 is orthogonal to L_α , $\forall \alpha \in \Phi$ by the previous proposition. If $z \in L_0$ is orthogonal to L_0 , then $\kappa(z, L) = 0 \rightarrow z = 0$.

Proposition 2.2. Let H be a maximal toral subalgebra of L . Then $H = C_L(H)$.

pf. Denote $C = C_L(H)$.

- **C contains the semisimple and nilpotent parts of its elements.**

pf. $x \in C_L(H)$ then adx maps the subspace of H of L into the subspace 0. Then $\begin{cases} (\text{adx})_s \\ (\text{adx})_n \end{cases}$ have the same property by the proposition 1.4. Also $\begin{cases} (\text{adx})_s = \text{adx}_s \\ (\text{adx})_n = \text{adx}_n \end{cases}$, so the claim follows.

- **All semisimple elements of C lie in H .**

pf. If x is semisimple and centralises H , $H + Fx$ is toral since the sum of commuting semisimple elements are again semisimple. By the maximality of H , $H + Fx = H \rightarrow x \in H$.

- **The restriction of κ to H is nondegenerate.**

pf. Let $\kappa(h, H) = 0$ for some $h \in H$. If $x \in C$ is nilpotent, then $\begin{cases} [xH] = 0 \\ \text{adx nilpotent} \end{cases}$ implies $T(\text{adxady}) = 0$, $\forall y \in H$ or $\kappa(x, H) = 0$ by the previous lemma. Then the previous statements imply $\kappa(h, C) = 0$ whence $h = 0$.

- **C is nilpotent.**

pf. If $x \in C$ is semisimple, $x \in H$ by the second statement, and $\text{ad}_C x = 0$ is certainly nilpotent. On the other hand, if $x \in C$ is nilpotent, then $\text{ad}_C x$ is a fortiori nilpotent. Now let $x \in C$ be arbitrary, $x = x_s + x_n$. Since $x_s, x_n \in C$ by the first statement, $\text{ad}_C x$ is the sum of commuting nilpotents and is therefore itself nilpotent. Then by Engel's theorem, C is nilpotent.

Since κ is associative and $[HC] = 0$, $\kappa(H, [CC]) = 0$. Then since the restriction of κ to H is nondegenerate, $H \cap [CC] = 0$. If C is not abelian, $[CC] \neq 0$. Then since C is nilpotent, $Z(C) \cap [CC] \neq 0$ by the lemma 1.2. Let $z \neq 0$ lie in this intersection. Then z cannot be semisimple. Its nilpotent part n is therefore nonzero and lies in C , hence also lies in $Z(C)$. Then the lemma implies $\kappa(n, C) = 0$, a contradiction to the previous corollary. So C is abelian. If $C \neq H$, C contains a nonzero nilpotent element x by the first and second statements. Then $\kappa(x, y) = T(\text{adxady}) = 0$, $\forall y \in C$, a contradiction to the previous corollary. So $C = H$ and the proposition follows.

Corollary 2.5. *The restriction of κ to H is nondegenerate.*

$$\rightarrow H \sim H^*$$

$$\forall \phi \in H^*, \exists ! t_\phi \in H \text{ s.t. } \phi(h) = \kappa(t_\phi, h), \forall h \in H.$$

Proposition 2.3. 1. Φ spans H^*

$$2. \text{ If } \alpha \in \Phi, -\alpha \in \Phi$$

$$3. \text{ Let } \alpha \in \Phi, x \in L_\alpha, y \in L_{-\alpha}. \text{ Then } [xy] = \kappa(x, y)t_\alpha$$

$$4. \text{ If } \alpha \in \Phi, [L_\alpha L_{-\alpha}] \text{ is one dimensional with basis } t_\alpha$$

$$5. \alpha(t_\alpha) = \kappa(t_\alpha, t_\alpha) \neq 0, \forall \alpha \in \Phi$$

$$6. \text{ If } \alpha \in \Phi, \text{ and } x_\alpha \text{ is any nonzero element of } L_\alpha, \text{ then there exists } y_\alpha \in L_{-\alpha} \text{ s.t. } x_\alpha, y_\alpha, h_\alpha = [x_\alpha y_\alpha] \text{ spans a three dimensional simple subalgebra of } L \text{ isomorphic to } \mathfrak{sl}(2, F) \text{ via } x_\alpha \mapsto \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, y_\alpha \mapsto \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, h_\alpha \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$7. h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$$

pf.

- If Φ fails to span H^* , by the duality, $\exists h \in H$ s.t. $\alpha(h) = 0, \forall \alpha \in \Phi$. Then $[hL_\alpha] = 0, \forall \alpha \in \Phi$. Since $[hH] = 0, [hL] = 0$, i.e. $h \in Z(L) = 0$ a contradiction.
- Let $\alpha \in \Phi$. If $-\alpha \notin \Phi$, $\kappa(L_\alpha, L_\beta) = 0, \forall \beta \in H^*$ by the proposition 2.1. Therefore $\kappa(L_\alpha, L) = 0$, a contradiction to the nondegeneracy of κ
- Let $\alpha \in \Phi, x \in L_\alpha, y \in L_{-\alpha}$. Let $h \in H$ be arbitrary. Then

$$\begin{aligned} \kappa(h, [xy]) &= \kappa([hx], y) && \text{by the associativity of } \kappa \\ &= \alpha(h)\kappa(x, y) && = \kappa(t_\alpha, h)\kappa(x, y) \\ &= \kappa(\kappa(x, y)t_\alpha, h) && \text{by the bilinearity of } \kappa \\ &= \kappa(h, \kappa(x, y)t_\alpha) && \text{by the symmetry of } \kappa \end{aligned} \Leftrightarrow H \perp [xy] - \kappa(x, y)t_\alpha \text{ and}$$
 therefore $[xy] = \kappa(x, y)t_\alpha$ by the previous corollary.
- By 3, t_α spans $[L_\alpha L_{-\alpha}]$, provided $[L_\alpha L_{-\alpha}] \neq 0$. Let $0 \neq x \in L_\alpha$. If $\kappa(x, L_{-\alpha}) = 0, \kappa(x, L) = 0$ a contradiction to the nondegeneracy of κ . Thus $\exists 0 \neq y \in L_{-\alpha}$ s.t. $\kappa(x, y) \neq 0$. Then $[xy] \neq 0$.
- Suppose the contrary. Then $\alpha(t_\alpha) = 0$ so that $[t_\alpha x] = 0 = [t_\alpha y], \forall x \in L_\alpha, \forall y \in L_{-\alpha}$. By 4, $\exists x \in L_\alpha, y \in L_{-\alpha}$ s.t. $\kappa(x, y) \neq 0$. By modifying scalar, we may assume $\kappa(x, y) = 1$. Then $[xy] = t_\alpha$ by 3. Then the subspace of L spanned by x, y, t_α is a three dimensional solvable algebra, $S \cong \text{ad}_L S \subset \mathfrak{gl}(L)$. In particular $\text{ad}_L S$ is nilpotent $\forall s \in [SS]$. So $\text{ad}_L t_\alpha$ is both semisimple and nilpotent, i.e. $\text{ad}_L t_\alpha = 0$. This implies $t_\alpha \in Z(L) = 0$, a contradiction. Thus $\alpha(t_\alpha) \neq 0$.
- Given $x_\alpha \in L_\alpha, \exists y \in L_{-\alpha}$ s.t. $\kappa(x_\alpha, y_\alpha) = \frac{2}{\kappa(t_\alpha, t_\alpha)}$ by 5 and the fact that $\kappa(x_\alpha, L_{-\alpha}) \neq 0$. Set $h_\alpha = \frac{2t_\alpha}{\kappa(t_\alpha, t_\alpha)}$. Then $[x_\alpha y_\alpha] = h_\alpha$ by 3. Also $[h_\alpha x_\alpha] = \frac{2}{\alpha(t_\alpha)}[t_\alpha x_\alpha] = \frac{2\alpha(t_\alpha)}{\alpha(t_\alpha)}x_\alpha = 2x_\alpha$ and similarly $[h_\alpha y_\alpha] = -2y_\alpha$. So $x_\alpha, y_\alpha, h_\alpha$ span a three dimensional subalgebra of L with the same multiplication table as $\mathfrak{sl}(2, F)$.
- $\kappa(t_\alpha, h) = \alpha(h)$ implies $t_\alpha = -t_{-\alpha}$.

For each pair of roots $\pm\alpha$, let $S_\alpha \cong \mathfrak{sl}(2, F)$ be a subalgebra of L constructed as in the previous proposition. Then by Weyl's theorem and theorem 2.6., we have a complete description of all finite dimensional S_α -modules.

Fix $\alpha \in \Phi$. Consider the subspace M of L spanned by H along with all root spaces of the form $L_{c\alpha}, c \in F^*$. This is an S_α submodule of L by the proposition 2.1.

- Proposition 2.4.** 1. $\alpha \in \Phi$ implies $\dim L_\alpha = 1$. In particular, $S_\alpha = L_\alpha + L_{-\alpha} + H$ and for given nonzero $x_\alpha \in L_\alpha$, there exists a unique $y_\alpha \in L_{-\alpha}$ s.t. $[x_\alpha y_\alpha] = h_\alpha$.
2. if $\alpha \in \Phi$, the only scalar multiples of α which are roots are $\pm\alpha$

3. If $\alpha, \beta \in \Phi$, $\beta(h_\alpha) \in \mathbb{Z}$ and $\beta - \beta(h_\alpha)\alpha \in \Phi$. Here $\beta(h_\alpha)$ are called Cartan integers.
4. If $\alpha, \beta, \alpha + \beta \in \Phi$, $[L_\alpha L_\beta] = L_{\alpha+\beta}$
5. Let $\alpha, \beta \in \Phi$ s.t. $\beta \neq \pm\alpha$. Let r, q be the largest integers for which $\beta - r\alpha, \beta + q\alpha$ are roots respectively. Then all $\beta + i\alpha \in \Phi$ and $\beta(h_\alpha) = r - q$.

pf.

- 1,2

The weights of h_α on M are integers 0 and $2c = c\alpha(h_\alpha)$. In particular, all c occurring here must be integral multiples of $1/2$. Given $\ker \alpha \subset H$, S_α acts trivially on $\ker \alpha$, a subspace of codimension 1 in H complementary to Fh_α , while on the other hand S_α is itself an irreducible S_α -submodule of M . Thus $\ker \alpha, S_\alpha$ exhaust the occurrences of the weight 0 for h_α . Therefore the only even weights occurring in M are $0, \pm 2$. This proves 2α is not a root, i.e. twice a root is never a root. But then $\frac{\alpha}{2}$ cannot be a root either, so 1 cannot occur as a weight of h_α in M . Since $M = H + S_\alpha$ by the corollary 2.3., $\dim L_\alpha = 1$ and the only multiples of a root α which are roots are $\pm\alpha$.

- 3,4

Let $\beta \neq \pm\alpha$. Set $K = \sum_{i \in \mathbb{Z}} L_{\beta+i\alpha}$. Then each root space is one dimensional and no $\beta + i\alpha$ can equal 0. So K is an S_α -submodule of L , with one dimensional weight spaces for the distinct integral weights $\beta(h_\alpha) + 2i$ where $\beta + i\alpha \in \Phi$, $i \in \mathbb{Z}$. Not both 0 and 1 can occur as weights of this form so by the corollary 2.3., K is irreducible. The highest weight must be $\beta(h_\alpha) + 2q$ and the lowest must be $\beta(h_\alpha) - 2r$ if q, r is the largest integer for which $\beta + q\alpha$ or $\beta - r\alpha$ respectively is a root. Moreover, the weights on K form an arithmetic progression with difference 2 by the theorem 2.6., and therefore roots $\beta + i\alpha$ form a string $\beta - r\alpha, \dots, \beta, \dots, \beta + q\alpha$. Then $(\beta - r\alpha)(h_\alpha) = -(\beta + q\alpha)(h_\alpha) \Leftrightarrow \beta(h_\alpha) = r - q$. If $\alpha, \beta, \alpha + \beta \in \Phi$, then $\text{ad} L_\alpha$ maps L_β onto $L_{\alpha+\beta}$, i.e. $[L_\alpha L_\beta] = L_{\alpha+\beta}$.

Here bracket of two nilpotents $[x_\alpha y_\alpha] = h_\alpha$ gives the semisimple element. We can view nilpotent elements as shifting weights. In particular, here x_α shifts weight by α , while y_α shifts weight by $-\alpha$. Since by the proposition 2.1, $[L_\alpha L_\beta] \subset L_{\alpha+\beta}$, $[x_\alpha y_\alpha] \in L_{\alpha-\alpha} = L_0$, the abelian subalgebra where the weight is fixed.

Let L be a semisimple Lie algebra over the algebraically closed field F of characteristic 0, H be a maximal toral subalgebra, $\Phi \subset H^*$ be the set of roots of L relative to H and $L = H \oplus \bigoplus_{\alpha \in \Phi} L_\alpha$ be the root space decomposition. Since the

restriction to H of the Killing form is nondegenerate, we may transfer the form to H^* , letting $(\gamma, \delta) = \kappa(t_\gamma, t_\delta)$, $\forall \delta, \gamma \in H^*$. Since Φ spans H^* , choose a basis $\alpha_1, \dots, \alpha_l$ of H^* consisting of roots. If $\beta \in \Phi$, $\beta = \sum_{i=1}^l c_i \alpha_i$ for some $c_i \in F$.

CLAIM : $c_i \in \mathbb{Q}$

pf. $\forall j$, $(\beta, \alpha_j) = \sum_{i=1}^l c_i (\alpha_i, \alpha_j)$, so multiplying both sides by $\frac{2}{(\alpha_j, \alpha_j)}$ yields $\frac{2(\beta, \alpha_j)}{(\alpha_j, \alpha_j)} = \sum_i \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)} c_i$, a system of l equations in l unknowns c_i with integral coefficients. Since $(\alpha_1, \dots, \alpha_l)$ is a basis of H^* , and the form is nondegenerate, the matrix $((\alpha_i, \alpha_j))$ is nonsingular. So the same is true for the coefficient matrix of this system of equations. So the equations possess a unique solution over $\mathbb{Q}$.

$\rightarrow \mathbb{Q}$ -subspace $E_{\mathbb{Q}}$ of H^* spanned by all the roots has $\mathbb{Q}$ -dimension $l = \dim_F H^*$. Also $\forall \lambda, \mu \in H^*$, $(\lambda, \mu) = \kappa(t_\lambda, t_\mu) = \sum \alpha(t_\lambda) \alpha(t_\mu) = \sum (\alpha, \lambda)(\alpha, \mu)$ summing over $\alpha \in \Phi$. In particular, for $\beta \in \Phi$, $(\beta, \beta) = \sum (\alpha, \beta)^2$. So dividing both sides by $(\beta, \beta)^2$ gives $\frac{1}{(\beta, \beta)} = \sum \frac{(\alpha, \beta)^2}{(\beta, \beta)^2} = \frac{1}{4} \sum \left(\frac{2(\alpha, \beta)}{(\beta, \beta)} \right)^2 \in \mathbb{Q}$ given $\frac{2(\alpha, \beta)}{(\beta, \beta)} \in \mathbb{Z}$ by the previous proposition 3. Therefore $(\beta, \beta) \in \mathbb{Q}$, $(\alpha, \beta) \in \mathbb{Q}$. Thus all inner products of vectors in $E_{\mathbb{Q}}$ are rational, so we get a nondegenerate form on $E_{\mathbb{Q}}$. Also $(\lambda, \lambda) = \sum (\alpha, \lambda)^2$ gives that the form on $E_{\mathbb{Q}}$ is positive definite.

3 Root System

3.1 Definition

Consider a fixed Euclidean space E , i.e. a finite dimensional vector space over $\mathbb{R}$ endowed with a positive definite symmetric bilinear form (α, β) .

Definition 3.1. A reflection in E is an invertible linear transformation leaving pointwise fixed a hyperplane and sending any vector orthogonal to that hyperplane into its negative.

$\rightarrow$ a reflection is orthogonal in the sense that it preserves the inner product on E .

Each nonzero $\alpha \in E$ determines a reflection σ_α s.t. $\sigma_\alpha(\beta) = \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)} \alpha \equiv \beta - \langle \beta, \alpha \rangle \alpha$ with reflecting hyperplane $P_\alpha = \{\beta \in E \mid (\beta, \alpha) = 0\}$.

Lemma 3.1. Let Φ be a finite set which spans E . Suppose all reflections σ_α , $\forall \alpha \in \Phi$ leave Φ invariant. If $\sigma \in GL(E)$ leaves Φ invariant, fixes pointwise a hyperplane P of E and sends some nonzero $\alpha \in \Phi$ to its negative, then $\sigma = \sigma_\alpha$.

pf. Let $\tau = \sigma \sigma_\alpha = \sigma \sigma_\alpha^{-1}$. Then $\begin{cases} \tau(\Phi) = \Phi \\ \tau(\alpha) = \alpha \end{cases}$ and τ acts as the identity on $\begin{cases} \text{the subspace } \mathbb{R}\alpha \\ \text{the quotient } E/\mathbb{R}\alpha \end{cases}$. So all eigenvalues of τ are 1 and the minimal polyno-

mial of τ divides $(T-1)^l$ where $l = \dim E$. On the other hand, since Φ is finite, not all vectors $\beta, \tau(\beta), \dots, \tau^k(\beta)$ can be distinct, so some power of τ fixes β . Choose k large enough so that τ^k fixes all $\beta \in \Phi$. Because Φ spans E , $\tau^k = 1$ so the minimal polynomial of τ divides $T^k - 1$. In sum, τ has minimal polynomial $T - 1 = \gcd(T^k - 1, (T - 1)^l)$, i.e. $\tau = 1$ and the lemma follows.

Definition 3.2. A subset Φ of the Euclidean space E is called a root system in E if Φ is finite, spans E and $0 \notin \Phi$ the following axioms are satisfied $\left\{ \begin{array}{l} \text{if } \alpha \in \Phi, \text{ the only multiples of } \alpha \text{ in } \Phi \text{ are } \pm\alpha \\ \text{if } \alpha \in \Phi, \text{ the reflection } \sigma_\alpha \text{ leaves } \Phi \text{ invariant} \\ \text{if } \alpha, \beta \in \Phi, \langle \beta, \alpha \rangle \in \mathbb{Z} \end{array} \right.$

Let Φ be a root system in E . Denote by $\mathcal{W}$ the subgroup of $GL(E)$ generated by the reflections σ_α , $\forall \alpha \in \Phi$. Then by the third axiom in the definition, $\mathcal{W}$ permutes the set Φ . This allows us to identify $\mathcal{W}$ with a subgroup of the symmetric group on Φ given the fact that only the identity fixes all the roots by the first axiom; in particular, $\mathcal{W}$ is finite. $\mathcal{W}$ is called the **Weyl group** of Φ .

Lemma 3.2. Let Φ be a root system in E , with Weyl group $\mathcal{W}$. If $\sigma \in GL(E)$ leaves Φ invariant, then $\sigma\sigma_\alpha\sigma^{-1} = \sigma_{\sigma(\alpha)}$, $\forall \alpha \in \Phi$ and $\langle \beta, \alpha \rangle = \langle \sigma(\beta), \sigma(\alpha) \rangle$, $\forall \alpha, \beta \in \Phi$.

$$\begin{aligned} \sigma\sigma_\alpha\sigma^{-1}(\sigma(\beta)) &= \sigma\sigma_\alpha(\beta) \in \Phi && \text{since } \sigma_\alpha(\beta) \in \Phi \\ \text{pf.} &= \sigma(\beta - \langle \beta, \alpha \rangle \alpha) && \text{by the definition} \\ &= \sigma(\beta) - \langle \beta, \alpha \rangle \sigma(\alpha) \end{aligned} \quad . \quad \text{Since } \sigma(\beta) \text{ runs over } \Phi$$

as β runs over Φ , this implies $\sigma\sigma_\alpha\sigma^{-1}$ leaves Φ invariant while fixing pointwise the hyperplane $\sigma(P_\alpha)$ and sending $\sigma(\alpha)$ to $-\sigma(\alpha)$. Then by the previous lemma, $\sigma\sigma_\alpha\sigma^{-1} = \sigma_{\sigma(\alpha)}$. Then $\sigma_{\sigma(\alpha)}(\sigma(\beta)) = \sigma(\beta) - \langle \sigma(\beta), \sigma(\alpha) \rangle \sigma(\alpha) = \sigma(\beta) - \langle \beta, \alpha \rangle \sigma(\alpha)$ gives the last identity.

Definition 3.3. Call (Φ, E) , (Φ', E') isomorphic if there exists a vector space isomorphism $\phi: E \rightarrow E'$ sending Φ onto Φ' s.t. $\langle \phi(\beta), \phi(\alpha) \rangle = \langle \beta, \alpha \rangle$, $\forall \alpha, \beta \in \Phi$.

$\rightarrow \sigma_{\phi(\alpha)}(\phi(\beta)) = \phi(\sigma_\alpha(\beta))$ so there exists a natural isomorphism $\sigma \mapsto \phi \circ \sigma \circ \phi^{-1}$ of Weyl groups.

CLAIM : $\mathcal{W}$ is a normal subgroup of $\text{Aut}\Phi$, where $\text{Aut}\Phi$ is the group of automorphisms of Φ , i.e. isomorphisms of Φ onto itself.

pf. Since $\mathcal{W}$ is a subgroup of the symmetric group on Φ , and by the previous lemma, $\langle \sigma(\beta), \sigma(\alpha) \rangle = \langle \beta, \alpha \rangle$, $\mathcal{W} \subset \text{Aut}\Phi$. Also by the isomorphism defined in the previous stmt, it is a normal subgroup.

Definition 3.4. Define $\alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle}$. Call $\Phi^\vee = \{\alpha^\vee \mid \alpha \in \Phi\}$ the dual of Φ .

→ it is a root system in E whose Weyl group is isomorphic to $\mathcal{W}$.

pf. By the duality, the first and second axiom of the root system follows trivially.

CLAIM : $\langle \alpha^\vee, \beta^\vee \rangle = \frac{2(\alpha^\vee, \beta^\vee)}{(\beta^\vee, \beta^\vee)} = \frac{2(\frac{2\alpha}{(\alpha, \alpha)}, \frac{2\beta}{(\beta, \beta)})}{(\frac{2\beta}{(\beta, \beta)}, \frac{2\beta}{(\beta, \beta)})} = \frac{2(\alpha, \beta)}{(\alpha, \alpha)} = \langle \beta, \alpha \rangle$.

Thus, if $\alpha^\vee, \beta^\vee \in \Phi^\vee$, $\langle \beta^\vee, \alpha^\vee \rangle = \langle \alpha, \beta \rangle \in \mathbb{Z}$ since $\alpha, \beta \in \Phi$. Also, if $\alpha^\vee \in \Phi^\vee$, $\sigma_{\alpha^\vee}(\beta^\vee) = \beta^\vee - \langle \beta^\vee, \alpha^\vee \rangle \alpha^\vee = \frac{2\beta}{(\beta, \beta)} - \langle \alpha, \beta \rangle \frac{2\alpha}{(\alpha, \alpha)}$. Let $\gamma = \beta - \langle \beta, \alpha \rangle \alpha \in \Phi$. Then it equals $\beta - \langle \alpha^\vee, \beta^\vee \rangle \alpha$ and $\gamma^\vee = \frac{2\gamma}{(\gamma, \gamma)} = \frac{2\beta - 2\langle \beta, \alpha \rangle \alpha}{(\beta, \beta) + \langle \beta, \alpha \rangle^2 (\alpha, \alpha) - 2\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle} = \frac{2\beta - \frac{4(\beta, \alpha)}{(\alpha, \alpha)} \alpha}{(\beta, \beta) + 2\langle \beta, \alpha \rangle \langle \beta, \alpha \rangle - 2\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle} = \frac{2\beta}{(\beta, \beta)} - \frac{4(\beta, \alpha)}{(\alpha, \alpha)(\beta, \beta)} \alpha = \sigma_{\alpha^\vee}(\beta^\vee)$. Then it leaves $\Phi^\vee$ invariant. Thus it is a root system.

For all $\beta \in E$, $\sigma_{\alpha^\vee}(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \alpha^\vee = \beta - \frac{2(\beta, \alpha^\vee)}{(\alpha^\vee, \alpha^\vee)} \frac{2\alpha}{(\alpha, \alpha)} = \beta - \langle \beta, \alpha \rangle \alpha = \sigma_\alpha(\beta)$ and the Weyl group is isomorphic to $\mathcal{W}$.

Definition 3.5. Call $l = \dim E$ the rank of the root system Φ .

Given θ , the angle between vectors $\alpha, \beta \in E$, $\begin{cases} \langle \beta, \alpha \rangle = 2 \frac{\|\beta\|}{\|\alpha\|} \cos \theta \\ \langle \alpha, \beta \rangle \langle \beta, \alpha \rangle = 4 \cos^2 \theta \end{cases}$. This

narrows down the possibilities for the values for $\langle \alpha, \beta \rangle, \langle \beta, \alpha \rangle, \theta, (\frac{\|\beta\|}{\|\alpha\|})^2$. For non-proportional α, β with $\|\beta\| \geq \|\alpha\|$:

$\langle \alpha, \beta \rangle$	$\langle \beta, \alpha \rangle$	θ	$\ \beta\ ^2/\ \alpha\ ^2$
0	0	$\pi/2$	undetermined
1	1	$\pi/3$	1
-1	-1	$2\pi/3$	1
1	2	$\pi/4$	2
-1	-2	$3\pi/4$	2
1	3	$\pi/6$	3
-1	-3	$5\pi/6$	3

Lemma 3.3. Let α, β be nonproportional roots. If $(\alpha, \beta) > 0$, i.e. if the angle between α, β is strictly acute, then $\alpha - \beta$ is a root. If $(\alpha, \beta) < 0$, $\alpha + \beta$ is a root.

pf. For $(\alpha, \beta) > 0$, it is possible iff $\langle \alpha, \beta \rangle$ or $\langle \beta, \alpha \rangle$ equals 1 by the above table.

$\begin{cases} \langle \alpha, \beta \rangle = 1 \rightarrow \sigma_\beta(\alpha) = \alpha - \beta \in \Phi \\ \langle \beta, \alpha \rangle = 1 \rightarrow \beta - \alpha \in \Phi \rightarrow \sigma_{\beta-\alpha}(\beta - \alpha) = \alpha - \beta \in \Phi \end{cases}$. So the first part follows. The second follows by the first with replacing β with $-\beta$.

3.2 Simple Roots

Let Φ denote a root system of rank l in a Euclidean space E with Weyl group $\mathcal{W}$.

Definition 3.6. A subset Δ of Φ is called a base if

$\begin{cases} \Delta \text{ is a basis of } E \\ \text{each root } \beta = \sum_{\alpha \in \Delta} k_\alpha \alpha \text{ with integral coefficients } k_\alpha \text{ all nonnegative or nonpositive} \end{cases}$.
The roots in Δ are called simple.

→ define the height of a root by $\text{ht}\beta = \sum_{\alpha \in \Delta} k_{\alpha}$. If $\begin{cases} \forall k_{\alpha} \geq 0 \\ \forall k_{\alpha} \leq 0 \end{cases}$, β is called $\begin{cases} \text{positive} \\ \text{negative} \end{cases}$ and write $\begin{cases} \beta > 0 \\ \beta < 0 \end{cases}$. The collection of $\begin{cases} \text{positive} \\ \text{negative} \end{cases}$ roots is denoted $\begin{cases} \Phi^+ \\ \Phi^- \end{cases}$. If α, β are positive roots, and $\alpha + \beta$ is a root, it is also positive. Δ defines a partial order on E by $\beta < \alpha$ iff $\alpha - \beta$ is a sum of positive roots or $\beta = \alpha$.

Lemma 3.4. *If Δ is a base of Φ , then $(\alpha, \beta) \leq 0$, $\forall \alpha \neq \beta \in \Delta$ and $\alpha - \beta$ is not a root.*

pf. Assume the contrary. Then $(\alpha, \beta) > 0$. Since $\alpha \neq \beta$, by the hypothesis and $\alpha \neq -\beta$ trivially, by the previous lemma, $\alpha - \beta$ is a root, a contradiction to the second condition in the definition of a base.

Definition 3.7. $\forall \gamma \in E$, define $\Phi^+(\gamma) = \{\alpha \in \Phi \mid (\gamma, \alpha) > 0\}$ = the set of roots lying on the positive side of the hyperplane orthogonal to γ . Call $\gamma \in E$ regular if $\gamma \in E \setminus \cup_{\alpha \in \Phi} P_{\alpha}$, singular otherwise.

→ if γ is regular, $\Phi = \Phi^+(\gamma) \cup -\Phi^+(\gamma)$.

Definition 3.8. Call $\alpha \in \Phi^+(\gamma)$ decomposable if $\alpha = \beta_1 + \beta_2$, $\beta_i \in \Phi^+(\gamma)$, indecomposable otherwise.

CLAIM : The intersection of positive open half-spaces associated with any basis $\gamma_1, \dots, \gamma_l$ of E is nonvoid.

pf. If δ_i is the projection of γ_i on the orthogonal complement of the subspace spanned by all basis vectors except γ_i , consider $\gamma = \sum r_i \delta_i$ with all $r_i > 0$. Here $\delta_i \perp \gamma_j$, $i \neq j$ so only one term survives which is positive. Then such γ belongs to the intersection of these half-spaces, which is therefore nonvoid.

Theorem 3.1. *Let $\gamma \in E$ be regular. Then the set $\Delta(\gamma)$ of all indecomposable roots in $\Phi^+(\gamma)$ is a base of Φ .*

pf.

- each root in $\Phi^+(\gamma)$ is a nonnegative $\mathbb{Z}$ -linear combination of $\Delta(\gamma)$.

pf. Assume the contrary. Then $\exists \alpha \in \Phi^+(\gamma)$ s.t. it cannot be written in such a way. Choose α so that (γ, α) is as small as possible. Then α cannot be in $\Delta(\gamma)$, so $\alpha = \beta_1 + \beta_2$, $\beta_i \in \Phi^+(\gamma)$ and therefore $(\gamma, \alpha) = (\gamma, \beta_1) + (\gamma, \beta_2)$. Then each of (γ, β_i) is positive, so β_1, β_2 must each be a nonnegative $\mathbb{Z}$ -linear combination of $\Delta(\gamma)$ by the minimality of (γ, α) , which implies so is α , a contradiction.

- if $\alpha, \beta \in \Delta(\gamma)$, $(\alpha, \beta) \leq 0$ unless $\alpha = \beta$.

pf. Assume the contrary. Then by the acute-angle lemma, $\alpha - \beta$ is a root, since β clearly cannot be $-\alpha$. So $\alpha - \beta$ or $\beta - \alpha$ is in $\Phi^+(\gamma)$. Then either $\alpha = \beta + (\alpha - \beta)$ is decomposable or $\beta = \alpha + (\beta - \alpha)$ is decomposable, a contradiction to $\alpha, \beta \in \Delta(\gamma)$.

- $\Delta(\gamma)$ is a linearly independent set.

pf. Suppose $\sum r_\alpha \alpha = 0$ for some $r_\alpha \in \mathbb{R}$. Separating the indices α for which $r_\alpha > 0$ from those for which $r_\alpha < 0$, we can rewrite $\sum s_\alpha \alpha = \sum t_\beta \beta$, $s_\alpha, t_\beta > 0$. Call $\epsilon = \sum s_\alpha \alpha$. Then $(\epsilon, \epsilon) = \sum_{\alpha, \beta} s_\alpha t_\beta (\alpha, \beta) \leq 0$ by the second stmt. This forces $\epsilon = 0$. Then $0 = (\gamma, \epsilon) = \sum s_\alpha (\gamma, \alpha)$ forcing $s_\alpha = 0$, $\forall \alpha$. Similarly, $t_\beta = 0$.

Since $\Phi = \Phi^+(\gamma) \cup -\Phi^+(\gamma)$, by the first stmt, it satisfies the second condition of the base and also implies it spans E . By the third stmt, it is linearly independent and therefore it is a basis for E and $\Delta(\gamma)$ is a base of Φ .

→ **each base Δ of Φ has the form $\Delta(\gamma)$ for some regular $\gamma \in E$.**

pf. Given Δ , select $\gamma \in E$ so that $(\gamma, \alpha) > 0$, $\forall \alpha \in \Delta$. This is possible since the intersection of positive open half-spaces associated with any basis of E is nonvoid.

Then γ is regular and $\begin{cases} \Phi^+ \subset \Phi^+(\gamma) \\ \Phi^- \subset -\Phi^+(\gamma) \end{cases}$ by the second condition in the definition of the base so equality must hold in each instance. Since $\Phi^+ = \Phi^+(\gamma)$, Δ consists of indecomposable elements, i.e. $\Delta \subset \Delta(\gamma)$. But $\text{Card} \Delta = \text{Card} \Delta(\gamma) = l$, so $\Delta = \Delta(\gamma)$.

Definition 3.9. *The connected components of $E \setminus \cup_{\alpha \in \Phi} P_\alpha$ are called the open Weyl chambers of E .*

→ each regular $\gamma \in E$ belongs to precisely one Weyl chamber, denoted $\mathcal{C}(\gamma)$.

Weyl chambers are in natural one to one correspondence with bases. ($\mathcal{C}(\gamma) = \mathcal{C}(\gamma')$ is equivalent to γ, γ' lie on the same side of each hyperplane $P_\alpha, \forall \alpha \in \Phi \Leftrightarrow \Phi^+(\gamma) = \Phi^+(\gamma') \Leftrightarrow \Delta(\gamma) = \Delta(\gamma')$). If $\Delta = \Delta(\gamma)$, $\mathcal{C}(\Delta) = \mathcal{C}(\gamma)$ is called the fundamental Weyl chamber relative to Δ . The Weyl group sends one Weyl chamber onto another, i.e. $\sigma(\mathcal{C}(\gamma)) = \mathcal{C}(\sigma\gamma)$, $\forall \sigma \in \mathcal{W}$, γ regular. On the other hand, $\mathcal{W}$ permutes bases, i.e. σ sends Δ to $\sigma(\Delta)$, which is again a base. So we have $\sigma(\Delta(\gamma)) = \Delta(\sigma\gamma)$ because $(\sigma\gamma, \sigma\alpha) = (\gamma, \alpha)$.

CLAIM : Let $\Phi^\vee$ be the dual system of Φ , $\Delta^\vee = \{\alpha^\vee | \alpha \in \Delta\}$. Then $\Delta^\vee$ is a base of $\Phi^\vee$.

pf. Let $\gamma \in E$ be regular, belong to the Weyl chamber $\mathcal{C}(\gamma)$. Then $\gamma^\vee$ belongs to

the Weyl chamber $\mathcal{C}(\gamma^\vee)$, i.e. there is a one to one correspondence between Weyl chamber of Φ to ones of $\Phi^\vee$. Assume that $\exists \beta \in \Delta = \Delta(\gamma)$ s.t. $\beta^\vee$ not simple. Then by the lemma 3.5., $\exists \delta$ in the base of $\Phi^\vee$ s.t. $\beta^\vee - \delta$ is a positive root of $\Phi^\vee$. By duality, $\exists \mu \in \Phi$ s.t. $\frac{2\mu}{(\mu, \mu)} = \delta$. This implies $\frac{2\beta}{(\beta, \beta)} - \frac{2\mu}{(\mu, \mu)}$ is a root of $\Phi^\vee$ hence has all nonnegative coefficients for simple roots $\nu \in \Delta$ by the chamber correspondence. This is only possible when β has simple roots which coincide with μ which is impossible given β simple. Hence the claim follows.

Let Δ be a fixed base of Φ .

Lemma 3.5. *If α is positive but not simple, then $\alpha - \beta$ is a positive root for some $\beta \in \Delta$.*

pf. If $(\alpha, \beta) \leq 0$, $\forall \beta \in \Delta$, $\Delta \cup \{\alpha\}$ is a linearly independent set (set $\Delta' = \Delta \cup \{\alpha\}$ and $\sum_{\gamma \in \Delta'} k_\gamma \gamma = 0$. Then we can separate γ whose coefficient is positive from ones with negative coefficients. Then inner product gives all those coefficients should be zero, which implies linear independence), a contradiction to the definition of Δ . So $\exists \beta \in \Delta$ s.t. $(\alpha, \beta) > 0$ and then by the acute-angle lemma $\alpha - \beta \in \Phi$. Write $\alpha = \sum_{\gamma \in \Delta} k_\gamma \gamma$ with all $k_\gamma \geq 0$ and at least one $\gamma \neq \beta$ with $k_\gamma > 0$. Subtracting β from α yields a $\mathbb{Z}$ -linear combination of simple roots with at least one positive coefficient. This forces all coefficients to be nonnegative by the uniqueness of the expression.

Corollary 3.1. *Each $\beta \in \Phi^+$ can be written in the form $\alpha_1 + \dots + \alpha_k$ with $\alpha_i \in \Delta$ (not necessarily distinct) in such a way that each partial sum $\alpha_1 + \dots + \alpha_i$ is a root.*

pf. When $\text{ht}\beta = 1$, β itself is in Δ and there is nothing to prove, When $\text{ht}\beta > 1$, since it is positive root and not simple, by the previous lemma, $\exists \beta' \in \Delta$ s.t. $\beta - \beta'$ is a positive root. Then $\text{ht}(\beta - \beta') < \text{ht}\beta$ and therefore by the inductive hypothesis, it can be written as a sum of simple roots s.t. its partial sum is also a root. Then $\beta = \beta' + \text{such sum}$ gives the desired form.

Lemma 3.6. *Let α be simple. Then σ_α permutes the positive roots other than α .*

pf. Let $\beta \in \Phi^+ \setminus \{\alpha\}$, $\beta = \sum_{\gamma \in \Delta} k_\gamma \gamma$, $k_\gamma \in \mathbb{Z}_{\geq 0}$. Then β is not proportional to α . Therefore $k_\gamma \neq 0$, for some $\gamma \neq \alpha$. But the coefficient of γ in $\sigma_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha$ is still k_γ , i.e. $\sigma_\alpha(\beta)$ has at least one positive coefficient relative to Δ , forcing it to be positive. Moreover, $\sigma_\alpha(\beta) \neq \alpha$ since α is the image of $-\alpha$.

Corollary 3.2. *Set $\delta = \frac{1}{2} \sum_{\beta > 0} \beta$. Then $\sigma_\alpha(\delta) = \delta - \alpha$, $\forall \alpha \in \Delta$.*

pf. By the previous lemma, σ_α permutes all positive roots other than α so the sum of the positive roots other than α is left invariant. Since α is sent to $-\alpha$, the corollary follows.

Lemma 3.7. *Let $\alpha_1, \dots, \alpha_t \in \Delta$. Write $\sigma_i = \sigma_{\alpha_i}$. If $\sigma_1 \dots \sigma_{t-1}(\alpha_t)$ is negative, then for some index $1 \leq s < t$, $\sigma_1 \dots \sigma_t = \sigma_1 \dots \sigma_{s-1} \sigma_{s+1} \dots \sigma_{t-1}$.*

pf. Write $\beta_i = \begin{cases} \sigma_{i+1} \dots \sigma_{t-1}(\alpha_t), & 0 \leq i \leq t-2 \\ \alpha_t, & i = t-1 \end{cases}$. Since $\beta_0 < 0$, $0 < \beta_{t-1}$, we can find the smallest index s for which $0 < \beta_s$. Then $\sigma_s(\beta_s) = \beta_{s-1} < 0$ and by the previous lemma $\beta_s = \alpha_s$. Then since $\sigma \in \mathcal{W}$ implies $\sigma_{\sigma(\alpha)} = \sigma \sigma_\alpha \sigma^{-1}$, $\sigma_s = (\sigma_{s+1} \dots \sigma_{t-1}) \sigma_t (\sigma_{t-1} \dots \sigma_{s+1})$ which yields the lemma.

Corollary 3.3. *If $\sigma = \sigma_1 \dots \sigma_t$ is an expression for $\sigma \in \mathcal{W}$ in terms of reflections corresponding to simple roots, with t as small as possible, then $\sigma(\alpha_t) < 0$.*

pf. The minimality of t , together with the previous lemma, gives the corollary.

Theorem 3.2. 1. *If $\gamma \in E$ regular, there exists $\sigma \in \mathcal{W}$ s.t. $(\sigma(\gamma), \alpha) > 0$, $\forall \alpha \in \Delta$*

2. *If Δ' is another base of Φ , then $\sigma(\Delta') = \Delta$ for some $\sigma \in \mathcal{W}$*

3. *If α is any root, there exists $\sigma \in \mathcal{W}$ s.t. $\sigma(\alpha) \in \Delta$*

4. *$\mathcal{W}$ is generated by the σ_α , $\forall \alpha \in \Delta$*

5. *If $\sigma(\Delta) = \Delta$, $\sigma \in \mathcal{W}$, then $\sigma = 1$.*

pf. Define $\mathcal{W}'$ a subgroup of $\mathcal{W}$ generated by all σ_α , $\alpha \in \Delta$.

- Write $\delta = \frac{1}{2} \sum_{0 < \alpha} \alpha$ and choose $\sigma \in \mathcal{W}'$ for which $(\sigma(\gamma), \delta)$ is as big as possible. If α is simple, $\sigma_\alpha \sigma \in \mathcal{W}'$ so the choice of σ implies $(\sigma(\gamma), \delta) \geq (\sigma_\alpha \sigma(\gamma), \delta) = (\sigma(\gamma), \sigma_\alpha(\delta)) = (\sigma(\gamma), \delta - \alpha)$ by the corollary 3.2. The right side equals $(\sigma(\gamma), \delta) - (\sigma(\gamma), \alpha)$ which implies $(\sigma(\gamma), \alpha) \geq 0$, $\forall \alpha \in \Delta$. But $(\sigma(\gamma), \alpha) \neq 0$ since otherwise γ would be orthogonal to $\sigma^{-1}\alpha$. So all the inequalities are strict. Therefore, $\sigma(\gamma)$ lies in the fundamental Weyl chamber $\mathcal{C}(\Delta)$ and σ sends $\mathcal{C}(\gamma)$ to $\mathcal{C}(\Delta)$ as desired.
- Since $\mathcal{W}'$ permutes the Weyl chambers transitively by 1, it also permutes the bases of Φ transitively.
- Since the only roots proportional to α are $\pm\alpha$, the hyperplanes P_β , $\beta \neq \pm\alpha$ are distinct from P_α so there exists $\gamma \in P_\alpha$, $\gamma \notin P_\beta$. Choose γ' close enough to γ so that $(\gamma', \alpha) = \epsilon > 0$ while $|(\gamma', \beta)| > \epsilon$, $\forall \beta \neq \pm\alpha$. Then α belongs to the base $\Delta(\gamma')$ since ϵ is the smallest value $(\gamma', \cdot)$ takes on positive roots, so α cannot be decomposable. So each root belongs to at least one base, and therefore by 2, 3 follows.
- By 3, find $\sigma \in \mathcal{W}'$ s.t. $\beta = \sigma(\alpha) \in \Delta$. Then $\sigma_\beta = \sigma_{\sigma(\alpha)} = \sigma \sigma_\alpha \sigma^{-1}$ so $\sigma_\alpha = \sigma^{-1} \sigma_\beta \sigma \in \mathcal{W}'$, i.e. $\mathcal{W} = \mathcal{W}'$.

- Let $\sigma(\Delta) = \Delta$, but $\sigma \neq 1$. If σ is written minimally as a product of one or more simple reflections, then by the corollary 3.3., a contradiction. So $\sigma = 1$.

When $\sigma \in \mathcal{W}$ is written as $\sigma_{\alpha_1} \dots \sigma_{\alpha_t}$, $\alpha_i \in \Delta$, t minimal, we call the expression **reduced** and write $l(\sigma) = t$, which is the **length** of σ relative to Δ . By definition, $l(1) = 0$.

Define $n(\sigma) =$ number of positive roots α for which $\sigma(\alpha) < 0$.

Lemma 3.8. $\forall \sigma \in \mathcal{W}$, $l(\sigma) = n(\sigma)$.

pf. When $l(\sigma) = 0$, lemma follows trivially since $l(\sigma) = 0$ implies $\sigma = 1$.
When $l(\sigma) > 0$, write σ in reduced form as $\sigma = \sigma_{\alpha_1} \dots \sigma_{\alpha_t}$ and set $\alpha = \alpha_t$. By the corollary 3.3., $\sigma(\alpha) < 0$. Then by the lemma 3.6., $n(\sigma\sigma_\alpha) = n(\sigma) - 1$. On the other hand, $l(\sigma\sigma_\alpha) = l(\sigma) - 1 < l(\sigma)$ so by the inductive hypothesis, $l(\sigma\sigma_\alpha) = n(\sigma\sigma_\alpha)$. Then $l(\sigma) = n(\sigma)$.

The closure of the fundamental Weyl chamber $\overline{\mathcal{C}(\Delta)}$ relative to Δ is a fundamental domain for the action of $\mathcal{W}$ on E , i.e. each vector in E is $\mathcal{W}$ -conjugate to precisely one point of this set.

Lemma 3.9. Let $\lambda, \mu \in \overline{\mathcal{C}(\Delta)}$. If $\sigma\lambda = \mu$ for some $\sigma \in \mathcal{W}$, then σ is a product of simple reflections which fix λ ; in particular, $\lambda = \mu$.

pf. When $l(\sigma) = 0$, $\sigma = 1$ and the lemma trivially holds.
When $l(\sigma) > 0$, by the previous lemma, σ must send some positive root to a negative root; so σ cannot send all simple roots to positive roots. Say $\sigma\alpha < 0$, $\alpha \in \Delta$. Then $0 \geq (\mu, \sigma\alpha) = (\sigma^{-1}\mu, \alpha) = (\lambda, \alpha) \geq 0$, $\lambda, \mu \in \overline{\mathcal{C}(\Delta)}$. This forces $(\lambda, \alpha) = 0$, $\sigma_\alpha\lambda = \lambda$, hence $(\sigma\sigma_\alpha)\lambda = \mu$. Then by the lemma 3.6. and 3.8., $l(\sigma\sigma_\alpha) = l(\sigma) - 1$ so the inductive hypothesis can be applied and get $\sigma\sigma_\alpha$ is a product of simple reflections fixing λ , so $\sigma = (\sigma\sigma_\alpha)\sigma_\alpha$ is too. Thus $\lambda = \mu$.

Φ is called irreducible if it cannot be partitioned into the union of two proper subsets s.t. each root in one set is orthogonal to each root in the other.

CLAIM : Φ is irreducible iff Δ cannot be partitioned in the way stated above. pf.

- Let $\Phi = \Phi_1 \cup \Phi_2$ with $(\Phi_1, \Phi_2) = 0$. Unless Δ is wholly contained in one factor, this induces a similar partition of Δ . But $\Delta \subset \Phi_1$ implies $(\Delta, \Phi_2) = 0$, i.e. $(E, \Phi_2) = 0$ since Δ spans E , a contradiction. So the partition of Δ exists.
- Let Φ be irreducible, but $\Delta = \Delta_1 \cup \Delta_2$ with $(\Delta_1, \Delta_2) = 0$. Each root is conjugate to a simple root by the previous theorem, so $\Phi = \Phi_1 \cup \Phi_2$, Φ_i the set of

roots having a conjugate in Δ_i . Since $(\alpha, \beta) = 0$ implies $\sigma_\alpha \sigma_\beta = \sigma_\beta \sigma_\alpha$, and $\mathcal{W}$ is generated by σ_α , each root in Φ_i is obtained from one in Δ_i by adding or subtracting elements of Δ_i . Therefore Φ_i lies in the subspace E_i of E spanned by Δ_i and $(\Phi_1, \Phi_2) = 0$. This forces either Φ_1 or Φ_2 equals $\emptyset$, i.e. $\Delta_1 = \emptyset$ or $\Delta_2 = \emptyset$.

CLAIM : Let Φ be irreducible. Then $\Phi^\vee$ is also irreducible. Also, if Φ has all roots of equal length, so does $\Phi^\vee$ and $\Phi^\vee \cong \Phi$.

pf. If $\Phi^\vee$ is reducible, $\Phi^\vee = \Phi_1^\vee \cup \Phi_2^\vee$, s.t. $(\Phi_1^\vee, \Phi_2^\vee) = 0$. Then $\Delta^\vee = \Delta_1^\vee \cup \Delta_2^\vee$ with $(\Delta_1^\vee, \Delta_2^\vee) = 0$. Since $\alpha^\vee$ is a positive multiple of α , this in turn implies $(\Delta_1, \Delta_2) = 0$, a contradiction to the irreducibility of Δ and therefore Φ .

By the definition of the length, $(\alpha^\vee, \alpha^\vee) = \frac{4}{(\alpha, \alpha)} = \frac{4}{c}$ where $c = (\alpha, \alpha)$ gives that if Φ has all roots of equal length, so does $\Phi^\vee$.

With all lengths equal $\langle \alpha, \beta \rangle = \langle \beta, \alpha \rangle$, so the Cartan matrix equals its transpose, which is the Cartan matrix of $\Phi^\vee$. Thus $\Phi^\vee \cong \Phi$ by Proposition 3.1 below.

Lemma 3.10. *Let Φ be irreducible. Relative to the partial ordering $<$, there is a unique maximal root β . Moreover, if $\beta = \sum_{\alpha \in \Delta} k_\alpha \alpha$, then all $k_\alpha > 0$.*

pf. Let $\beta = \sum k_\alpha \alpha$ be a maximal root in the ordering. Then $0 < \beta$. If $\begin{cases} \Delta_1 = \{\alpha \in \Delta | k_\alpha > 0\} \\ \Delta_2 = \{\alpha \in \Delta | k_\alpha = 0\} \end{cases}$,

then $\Delta = \Delta_1 \cup \Delta_2$ is a partition. Suppose Δ_2 is nonvoid. Then $(\alpha, \beta) \leq 0$, $\forall \alpha \in \Delta_2$ by the lemma 3.4. Then, since Φ is irreducible, at least one $\alpha \in \Delta_2$ must be nonorthogonal to Δ_1 , forcing $(\alpha, \alpha') < 0$ for some $\alpha' \in \Delta_1$. This implies $(\alpha, \beta) = \sum_{\alpha' \in \Delta_1} k_{\alpha'} (\alpha, \alpha') \leq 0$ and at least one term < 0 and therefore $(\alpha, \beta) < 0$ and $\alpha + \beta$ is a root by the acute-angle lemma, a contradiction to the maximality of β . Therefore, Δ_2 is empty and all $k_\alpha > 0$. Also, by the acute angle lemma, $(\alpha, \beta) \geq 0$, $\forall \alpha \in \Delta$ with $(\alpha, \beta) > 0$ for at least one α , since Δ spans E . Now let β' be another maximal root. Then by the previous argument, β' involves at least one $\alpha \in \Delta$ for which $(\alpha, \beta) > 0$. So $(\beta', \beta) > 0$ and $\beta - \beta'$ is a root by the acute-angle lemma unless $\beta = \beta'$. But if $\beta - \beta'$ is a root, either $\beta < \beta'$ or $\beta' < \beta$, a contradiction. So β is unique.

Lemma 3.11. *Let Φ be irreducible. Then $\mathcal{W}$ acts irreducibly on E . In particular, the $\mathcal{W}$ -orbit of a root α spans E .*

pf. Let E' be a nonzero subspace of E invariant under $\mathcal{W}$. The orthogonal complement E'' of E' is also $\mathcal{W}$ -invariant, and $E = E' \oplus E''$. Then $\forall \alpha \in \Phi$ either $\alpha \in E'$ or $E' \subset P_\alpha$ since $\sigma_\alpha(E') = E'$. Thus $\alpha \notin E'$ implies $\alpha \in E''$, so each root lies in one subspace or the other. This partitions Φ into orthogonal subsets, forcing one or the other to be empty. Since Φ spans E , $E' = E$.

The span of the $\mathcal{W}$ -orbit of a root is a nonzero $\mathcal{W}$ -invariant subspace of E , so the second stmt follows from the first.

Lemma 3.12. *Let Φ be irreducible. Then at most two root lengths occur in Φ and all roots of a given length are conjugate under $\mathcal{W}$.*

pf. If α, β are arbitrary roots, then not all $\sigma(\alpha)$, $\sigma \in \mathcal{W}$ can be orthogonal to β since $\sigma(\alpha)$ span E by the previous lemma. If $(\alpha, \beta) \neq 0$, the possible ratios of squared root lengths of α, β are $1, 2, 3, 1/2, 1/3$. These imply that at most two root lengths occur in Φ since the presence of three root lengths would yield also a ratio $3/2$.

Let α, β have equal length. After replacing one of these by $\mathcal{W}$ -conjugate, we may assume them to be nonorthogonal and distinct, otherwise there is nothing to prove. Then $\langle \alpha, \beta \rangle = \langle \beta, \alpha \rangle = \pm 1$. Replacing β by $-\beta = \sigma_\beta(\beta)$ we may assume $\langle \alpha, \beta \rangle = 1$. Thus $(\sigma_\alpha \sigma_\beta \sigma_\alpha)(\beta) = \sigma_\alpha \sigma_\beta(\beta - \alpha) = \sigma_\alpha(-\beta - \alpha + \beta) = \alpha$.

→ when Φ is irreducible with two distinct root lengths, we speak of long and short roots.

Lemma 3.13. *Let Φ be irreducible, with two distinct root lengths. Then the maximal root β in the lemma 3.10. is long.*

pf. Let $\alpha \in \Phi$ be arbitrary. Then it suffices to show $(\beta, \beta) \geq (\alpha, \alpha)$, $\forall \alpha$. So we may replace α by a $\mathcal{W}$ -conjugate lying in the closure of the fundamental Weyl chamber relative to Δ . Since $0 < \beta - \alpha$ by the lemma 3.10., $(\gamma, \beta - \alpha) \geq 0$, $\forall \gamma \in \overline{\mathcal{C}(\Delta)}$. Then when $\gamma = \beta$, $\gamma = \alpha$, $(\beta, \beta) \geq (\beta, \alpha) \geq (\alpha, \alpha)$.

3.3 Classification

Denote Φ a root system of rank l , $\mathcal{W}$ its Weyl group, Δ , a base of Φ .

Definition 3.10. *For a fixed ordering $(\alpha_1, \dots, \alpha_l)$ of the simple roots, the matrix $(\langle \alpha_i, \alpha_j \rangle)$ is called the Cartan matrix of Φ , and its entries are called Cartan integers.*

$$\begin{array}{lcl} A_1 \times A_1 & \rightarrow & \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ \text{e.g.} & & A_2 \rightarrow \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \\ & & B_2 \rightarrow \begin{bmatrix} 2 & -2 \\ -1 & 2 \end{bmatrix} \\ & & G_2 \rightarrow \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \end{array} \cdot$$

Proposition 3.1. *Let $\Phi' \subset E'$ be another root system, with base $\Delta' = \{\alpha'_1, \dots, \alpha'_l\}$. If $\langle \alpha_i, \alpha_j \rangle = \langle \alpha'_i, \alpha'_j \rangle$, $\forall 1 \leq i, j \leq l$, then the bijection $\alpha_i \mapsto \alpha'_i$ extends uniquely to an isomorphism $\phi: E \rightarrow E'$ mapping Φ onto Φ' and satisfying $\langle \phi(\alpha), \phi(\beta) \rangle = \langle \alpha, \beta \rangle$, $\forall \alpha, \beta \in \Phi$. Therefore, the Cartan matrix of Φ determines Φ up to isomorphism.*

pf. Since $\begin{cases} \Delta \text{ is a basis of } E \\ \Delta' \text{ is a basis of } E' \end{cases}$, there is a unique vector space isomorphism $\phi : E \rightarrow E'$ sending α_i to α'_i , $\forall 1 \leq i \leq l$. If $\alpha, \beta \in \Delta$, by the hypothesis $\sigma_{\phi(\alpha)}(\phi(\beta)) = \sigma_{\alpha'}(\beta') = \beta' - \langle \beta', \alpha' \rangle \alpha' = \phi(\beta) - \langle \beta, \alpha \rangle \phi(\alpha) = \phi(\beta - \langle \beta, \alpha \rangle \alpha) = \phi(\sigma_\alpha(\beta))$, i.e. $\forall \alpha \in \Delta$,

$$\begin{array}{ccc} E & \xrightarrow{\phi} & E' \\ \downarrow \sigma_\alpha & & \downarrow \sigma_{\phi(\alpha)} \\ E & \xrightarrow{\phi} & E' \end{array} \text{ commutes. The respective Weyl groups } \mathcal{W}, \mathcal{W}' \text{ are generated by}$$

simple reflections by the previous theorem. Write $\beta = \sigma(\alpha)$ with $\sigma \in \mathcal{W}$, $\alpha \in \Delta$. Then $\phi(\beta) = (\phi \circ \sigma \circ \phi^{-1})(\phi(\alpha)) \in \Phi'$ and therefore ϕ maps Φ onto Φ' . Moreover, the formula for a reflection shows ϕ preserves all Cartan integers.

→ it is possible to recover Φ from the Cartan integers. For example, start with the roots of height 1, i.e. the simple roots. For any pair $\alpha_i \neq \alpha_j$, the integer r for the α_j -string through α_i is 0, i.e. $\alpha_i - \alpha_j$ is not a root by the lemma 3.4. So the integer $q = -\langle \alpha_i, \alpha_j \rangle$. This lets us write down all roots α of height 2, hence all integers $\langle \alpha, \alpha_j \rangle$. For each root α of height 2, the integer r for the α_j -string through α can be determined easily, since α_j can be subtracted at most once and then q is found because $r - q = \langle \alpha, \alpha_j \rangle$. By the corollary 3.1., all positive roots are eventually obtained if we repeat this process.

Define the Coxeter graph of Φ to be a graph having l vertices, the i th joined to the j th by $\langle \alpha_i, \alpha_j \rangle \langle \alpha_j, \alpha_i \rangle$ edges.

e.g. $\begin{array}{l} A_1 \times A_1 \quad \bullet \quad \bullet \\ A_2 \quad \bullet \text{---} \bullet \\ B_2 \quad \bullet \text{====} \bullet \\ G_2 \quad \bullet \text{=====} \bullet \end{array}$

The Coxeter graph determines the numbers $\langle \alpha_i, \alpha_j \rangle$ in case all roots have equal length, since then $\langle \alpha_i, \alpha_j \rangle = \langle \alpha_j, \alpha_i \rangle$. Whenever a double or triple edge occurs in the Coxeter graph of Φ , we can add an arrow pointing to the shorter of the two roots. This allows us to recover the Cartan integers; we call the resulting figure the **Dynkin diagram** of Φ .

e.g. $\begin{array}{l} B_2 \quad \bullet \xrightarrow{\quad} \bullet \\ \alpha_1 \quad \alpha_2 \\ G_2 \quad \bullet \xleftarrow{\quad} \bullet \\ \alpha_1 \quad \alpha_2 \end{array}$

Φ is irreducible iff Φ cannot be partitioned into two proper orthogonal subsets. Also Φ is irreducible iff its Coxeter graph is connected. In general, there will be a number of connected components of the Coxeter graph.

Proposition 3.2. Φ decomposes as the union of irreducible root systems Φ_i in subspaces E_i of E s.t. $E = E_1 \oplus \cdots \oplus E_t$ orthogonal direct sum.

pf. For the Coxeter graph for Φ , let $\Delta = \Delta_1 \cup \cdots \cup \Delta_t$ be the partition of Δ into

mutually orthogonal subsets, namely the one given by the connected components of the Coxeter graph. If E_i is the span of Δ_i , $E = E_1 \oplus \cdots \oplus E_t$ orthogonal direct sum. Since every root is $\mathcal{W}$ -conjugate to a simple root by the theorem 3.2, it is the case.

Moreover, the $\mathbb{Z}$ -linear combinations of Δ_i which are roots obviously form a root system in E_i , whose Weyl group is the restriction to E_i of the subgroup of $\mathcal{W}$, generated by all σ_α , $\alpha \in \Delta_i$. Each E_i is $\mathcal{W}$ -invariant since $\alpha \notin \Delta_i$ implies σ_α acts trivially on E_i , so each root lies in one of the E_i , i.e. $\Phi = \Phi_1 \cup \cdots \cup \Phi_t$.

Theorem 3.3. *If Φ is an irreducible root system of rank l , its Dynkin diagram is one of the following :*

$A_l, l \geq 1$	$\bullet_1 \text{---} \bullet_2 \cdots \bullet_{l-1} \text{---} \bullet_l$	$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$
$B_l, l \geq 2$	$\bullet_1 \text{---} \bullet_2 \cdots \bullet_{l-1} \rightrightarrows \bullet_l$	$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -2 \\ & & & -1 & 2 \end{pmatrix}$
$C_l, l \geq 3$	$\bullet_1 \text{---} \bullet_2 \cdots \bullet_{l-1} \leftrightsquigarrow \bullet_l$	$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 \\ & & & -2 & 2 \end{pmatrix}$
$D_l, l \geq 4$	$\bullet_1 \text{---} \bullet_2 \cdots \bullet_{l-2} \begin{matrix} \nearrow \bullet_{l-1} \\ \searrow \bullet_l \end{matrix}$	$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 & -1 & -1 \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{pmatrix}$
E_6	$\bullet_1 \text{---} \bullet_3 \text{---} \bullet_4 \text{---} \bullet_5 \text{---} \bullet_6$ $\quad \quad \quad \downarrow \bullet_2$	$\begin{pmatrix} 2 & & -1 & & \\ -1 & 2 & & -1 & \\ & 2 & 2 & -1 & \\ -1 & -1 & -1 & 2 & -1 \\ & & -1 & 2 & -1 \end{pmatrix}$
E_7	$\bullet_1 \text{---} \bullet_3 \text{---} \bullet_4 \text{---} \bullet_5 \text{---} \bullet_6 \text{---} \bullet_7$ $\quad \quad \quad \downarrow \bullet_2$	$\begin{pmatrix} 2 & & -1 & & \\ -1 & 2 & & -1 & \\ & 2 & 2 & -1 & \\ -1 & -1 & -1 & 2 & -1 \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$
E_8	$\bullet_1 \text{---} \bullet_3 \text{---} \bullet_4 \text{---} \bullet_5 \text{---} \bullet_6 \text{---} \bullet_7 \text{---} \bullet_8$ $\quad \quad \quad \downarrow \bullet_2$	$\begin{pmatrix} 2 & & -1 & & \\ -1 & 2 & & -1 & \\ & 2 & 2 & -1 & \\ -1 & -1 & -1 & 2 & -1 \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \\ & & & & -1 & 2 \end{pmatrix}$
F_4	$\bullet_1 \text{---} \bullet_2 \rightrightarrows \bullet_3 \text{---} \bullet_4$	$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -2 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{pmatrix}$
G_2	$\bullet_1 \leftrightsquigarrow \bullet_2$	$\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$

In the Cartan matrices, blank entries are 0.

pf. Assume E is a Euclidean space, $\mathcal{U} = \{\epsilon_1, \dots, \epsilon_n\}$ is a set of n linearly independent unit vectors which satisfy $\begin{cases} (\epsilon_i, \epsilon_j) \leq 0 \\ 4(\epsilon_i, \epsilon_j)^2 \in \{0, 1, 2, 3\} \end{cases}, i \neq j$. Such a set of vectors is called admissible. We attach a graph Γ to the set $\mathcal{U}$ with vertices i, j joined by $4(\epsilon_i, \epsilon_j)^2$ edges.

- If some of the ϵ_i are discarded, the remaining ones still form an admissible set, whose graph is obtained from Γ by omitting the corresponding vertices and all incident edges.
- The number of pairs of vertices in Γ connected by one edge is strictly less than n .

pf. Set $\epsilon = \sum_{i=1}^n \epsilon_i$. Since the ϵ_i are linearly independent, $\epsilon \neq 0$. So $0 < (\epsilon, \epsilon) = n + 2\sum_{i < j} (\epsilon_i, \epsilon_j)$. Let i, j be a pair of distinct indices for which $(\epsilon_i, \epsilon_j) \neq 0$. Then $4(\epsilon_i, \epsilon_j)^2 = 1, 2, 3$ so $2(\epsilon_i, \epsilon_j) \leq -1$. Then the number of such pairs cannot exceed $n - 1$.

- Γ contains no cycles.

pf. A cycle would be the graph Γ' of an admissible subset $\mathcal{U}'$ of $\mathcal{U}$ and then Γ' violates the second stmt with n replaced by $\text{Card } \mathcal{U}'$.

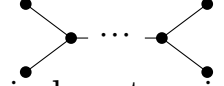
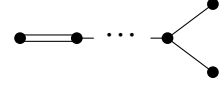
- No more than three edges can originate at a given vertex of Γ .

pf. Say $\epsilon \in \mathcal{U}$ and $\eta_1, \dots, \eta_k$ are the vectors in $\mathcal{U}$ connected to ϵ , i.e. $(\epsilon, \eta_i) < 0$ with $\epsilon, \eta_1, \dots, \eta_k$ all distinct. Then no two η 's can be connected so $(\eta_i, \eta_j) = 0, i \neq j$. Because $\mathcal{U}$ is linearly independent, some unit vector η_0 in the span of $\epsilon, \eta_1, \dots, \eta_k$ is orthogonal to $\eta_1, \dots, \eta_k$; clearly $(\epsilon, \eta_0) \neq 0$ for such η_0 . Then $\epsilon = \sum_{i=0}^k (\epsilon, \eta_i) \eta_i$, so $1 = (\epsilon, \epsilon) = \sum (\epsilon, \eta_i)^2$. This forces $\sum (\epsilon, \eta_i)^2 < 1 \Leftrightarrow \sum 4(\epsilon, \eta_i)^2 < 4$. But $4(\epsilon, \eta_i)^2$ is the number of edges joining ϵ to η_i in Γ .

- The only connected graph Γ of an admissible set $\mathcal{U}$ which can contain a triple edge is the Coxeter graph G_2 by the previous stmt.
- Let $\{\epsilon_1, \dots, \epsilon_k\} \subset \mathcal{U}$ have subgraph, a simple chain in Γ . If $\mathcal{U}' = (\mathcal{U} \setminus \{\epsilon_1, \dots, \epsilon_k\}) \cup \{\epsilon\}$, $\epsilon = \sum_{i=1}^k \epsilon_i$, then $\mathcal{U}'$ is admissible.

pf. Linear independence of $\mathcal{U}'$ is obvious. By hypothesis, $2(\epsilon_i, \epsilon_{i+1}) = -1, 1 \leq i \leq k-1$ so $(\epsilon, \epsilon) = k + 2\sum_{i < j} (\epsilon_i, \epsilon_j) = k - (k-1) = 1$. So ϵ is a unit vector. Any $\eta \in \mathcal{U} \setminus \{\epsilon_1, \dots, \epsilon_k\}$ can be connected to at most one of ϵ_i by the third stmt, so $(\eta, \epsilon) = 0$ or else $(\eta, \epsilon) = (\eta, \epsilon_i)$. In either case, $4(\eta, \epsilon)^2 \in \{0, 1, 2, 3\}$.

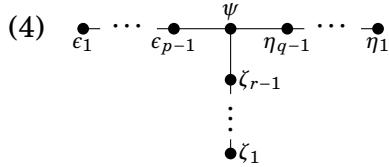
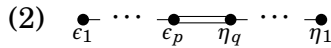
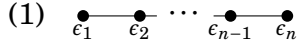
→ this excludes the subgraph of the form



since by the above stmt we replace the simple chain by a

single vertex, yielding graphs which violate the fourth stmt.

- Any connected graph Γ of an admissible set has one of the following forms:



pf. By the fifth stmt, we have the unique occurrence of triple edge. A connected graph containing more than one double edge would contain a subgraph which is not allowed by the previous stmt, so at most one double edge occurs. Moreover, if Γ has a double edge, it cannot also have a node by the previous stmt, so the second graph pictured is the only possibility. Finally, let Γ only have single edges; if Γ has no node, it must be a simple chain. It cannot contain more than one node, so the fourth graph is the only remaining possibility.

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